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of degree of **Master of Business Administration**
/Data Science

TITLE:

CUSTOMER SENTIMENT ANALYSIS ON AMAZON REVIEWS

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DECLARATION

I, Samudrala Komal Kumar, a student pursuing MBA / Data science at Amity University Online, hereby declare that the project work entitled “**Customer Sentiment Analysis on Amazon Reviews**” has been prepared by me during the academic year 2026 under the guidance of Vasanthi Chandran. I assert that this project is a piece of original bona-fide work done by me. It is the outcome of my own effort and that it has not been submitted to any other university for the award of any degree.

Signature of Student

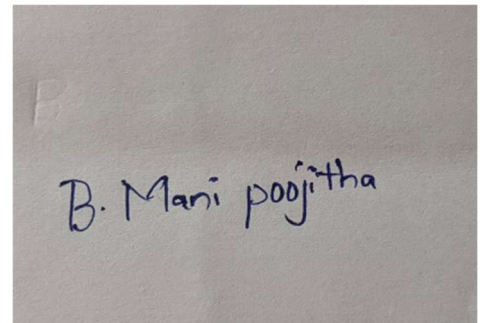
A photograph of a piece of paper with a handwritten signature in blue ink. The signature reads "B. Mani poojitha". There is a faint, embossed circular stamp in the background of the paper.

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<CHAPTER 1: INTRODUCTION TO THE TOPIC>

1.1 Introduction

As online shopping platforms have proliferated exponentially over the last few years, customer reviews have become one of the most far-reaching sources of information accessible to online consumers, companies, as well as the platforms themselves. Within a site as big and as varied as Amazon, millions of buyers post their reviews on a daily basis in an enormous array of product categories, creating a constant and ever-growing river of unstructured textual information that has massive analytical opportunities. Not only do the reviews allow the businesses to realise customer satisfaction on a fine grain level, but it also furnishes the raw material needed to bring significant improvement on the products and services as well as overall customer experience. The given project is devoted to sentiment analysis of the reviews of the musical instruments sold by Amazon on the one hand and, on the other hand, the classification of the expressed opinion of the users into three categories: positive, negative, or neutral attitudes. One of the most interesting areas of analysis, in this kind, is the musical instruments segment. Guitar, keyboard, drum and other equipment buyers are typically highly emotive about what they are buying music is often a highly emotional activity and not a strictly utilitarian affair and their reviews are thus more descriptive, emotional and full of subjective opinion. The feedback in this category often considers several dimensions at the same time, such as sound quality, the quality of building materials, playability, value-effort and purchasing experience itself which makes the sentiment classification task both challenging and analytically rewarding. In contrast to more traditional data analysis algorithms, which process structured numerical data, sentiment analysis uses Natural Language Processing and text mining methods to elicit subjective signals of written language in free-form. The difficulty is in fact in interpretation of this unstructured data, which is complicated by the broad range of language style, use of vocabulary, grammatical arrangement, usage of sarcasm and contextual meaning that exist between different reviewers and types of products. What seems to be an overwhelmingly positive review when inspecting word by word can actually express frustration or irony on the discourse level, and the only way to deal with such subtlety is to employ methods which consider both the semantic and syntactic levels of language as opposed to simple keyword matching. It has been developed in Java as the main language of programming,

demonstrating how the fundamental Java features, along with open-source NLP systems like Stanford CoreNLP and Apache OpenNLP can be efficiently used to create a workable text processing and sentiment classification system. Java is a language that is generally thought of in terms of enterprise-level software development and large-scale back-end systems, as opposed to data science apps, and the application of it to the NLP domain in this project shows that high-quality and useful text analytics are possible beyond the Python ecosystem where most modern machine learning applications are written. This project provides insight into a niche yet commercially viable and expanding segment of the e-commerce market by focusing on the musical instruments category on Amazon. The knowledge derived through systematic sentiment analysis in this regard, that is, determining the recurring quality problems, pricing problems, levels of performance satisfaction, and patterns of user experience in each product line (guitars, keyboards, percussion sets, etc.) have a pragmatic value to manufacturers seeking a direction to improve their products, sellers needing to refine their listing and positioning, and even the algorithmic recommendations and rankings used by Amazon itself. Moreover, the project can be used as an effective illustration that the divide between the old approach to software development and the latest text analytics can be successfully filled with the help of Java and create additional learning opportunities to students and developers interested in acquiring the skills of data science in a familiar and well-used programming language.

1.2 Objectives of the Work Undertaken

The ten objectives have been used to steer the project and they touch on the entire lifecycle of a sentiment analysis system, starting with data acquisition, all the way to implementation, evaluation, and future planning. All goals cover different and critical aspects of the work done.

1. To collect authentic customer reviews from Amazon's musical instruments category

The primary objective of the project will be to compile a list of authentic customer reviews that would include musical instruments like guitars, pianos, keyboards, and drums bought on Amazon through its review platform. This data is the raw material around which further analysis is based and should adequately represent customer opinions and experiences in real-world scenarios to facilitate useful classification and insight generation. The data collection procedure will include locating a publicly available and well-documented dataset in an established repository, examining its field structure and quality documentation, and loading

it into the development environment in a format that is easy to downstream preprocess and analyze.² To preprocess and clean unstructured review data to analyze it.

2. To clean and preprocess unstructured review data for analysis

Raw customer review data, as retrieved on online platforms, is always noisy and includes HTML markup, special characters, irrelevant symbols, duplicate data and inconsistencies in casing and formatting, which must first be removed before any meaningful analysis can be performed. The preprocessing goal involves a sequence of cleaning steps, such as HTML tag removal, text to lowercase conversion, punctuation and numeric token removal where they do not contain a sentiment signal, and the removal of common stop words which convert the raw input into a normalised, uniform, and analytically consumable format. Good preprocessing is a requirement of good classification and has been empirically demonstrated in the literature to be more effective than the selection of the classification algorithm itself on the ultimate model performance.³ To perform sentiment analysis with Java and NLP libraries.

3. To implement sentiment analysis using Java and NLP libraries

The technical deliverable of the project is to develop a working sentiment analysis engine in Java as the main language of implementation that builds on existing open-source NLP libraries such as Stanford CoreNLP and Apache OpenNLP. These libraries give the basic text processing functionality - such as tokenisation, part-of-speech tagging, lemmatisation, and sentiment scoring - needed to transform raw review text into formatted sentiment scores. Its use shows that complex natural language processing processes, typically identified with Python tools, can be successfully implemented within the Java ecosystem, with suitable library selection and well-designed software architecture.

4. To classify customer sentiments into positive, negative, or neutral categories

All reviews in the dataset will be automatically classified based on their vocalised tone of emotion, and each record will be allocated into one of three sentiment classes: positive, indicative of satisfaction and approval; negative, indicative of dissatisfaction, frustration, or criticism; and neutral, indicative of a neutral or neutral evaluation that is not strongly polarised in one direction or another. This classification step converts the unstructured review text into a structured categorical signal which can be aggregated, analysed and reported to the stakeholders in a format which can be easily read. The correctness of this classification is the main measure of the technical performance of the system and it is measured by standard measures such as precision, recall and the F1-score.

5. To use Natural Language Processing methods in Java. 5

To apply Natural Language Processing techniques using Java The project uses a variety of NLP techniques with the Java implementation to enhance the quality and richness of the sentiment classification beyond the basic text cleaning. They are tokenisation, that breaks down review text into word units to be analysed lexically; part-of-speech tagging, that determines the grammatical role of each word and helps to interpret it contextually; and lemmatisation, which normalises the vocabulary and reduces inflexed words to their dictionary root. Recognising named entities is also deemed as a possible optional addition to recognising product specific terms and brand mentions in the reviews. The combination of these methods allows the system to process language with the degree of structural awareness that extends beyond the word-frequency methods.

6. To evaluate the performance of the sentiment classification system

Strict assessment of the performance of the classification system is one of the fundamental goals, which will guarantee that the quality of the outputs of the model can be evaluated objectively and be compared with the set standards. Testing is done on a held-out set of reviews that was not utilized during system development, and a group of standard classification measures such as overall accuracy, per-class precision, per-class recall and the F1-score as the harmonic mean of the two. A confusion matrix is created to offer a deeper analysis of the correct and incorrect classifications in all three classes of sentiment, showing not just the overall accuracy of the system but also the specific strengths and limitations of the system in the classification of each category. This analysis stage determines what the system is doing well and the aspects that can be improved the most.

7. To visualize the sentiment analysis results for better understanding

Outputs of numerical classification are made more accessible and actionable by the production of data visualisations that convey sentiment distributions and trends in a more accessible, graphical format. Visualisation features of this project comprise pie charts of the proportionate distribution of positive, negative, and neutral reviews over the dataset; bar graphs of the sentiment distribution over the various products or time periods; word clouds of the most frequently used words in each sentiment category that would give a quick overview of the vocabulary that is most strongly linked to customer satisfaction and dissatisfaction. These visual deliverables are created with the help of charting features of JavaFX and are aimed at assisting non-technical stakeholders to interpret and take action on the results of the sentiment analysis.

8. To explore common themes and patterns in customer reviews

In addition to aggregate sentiment classification, the project will involve thematic analysis of the content of the reviews, to determine common themes, issues and compliments that describe customer experience with Amazon musical instruments. It can be found by reviewing how often certain product related terms, feature mentions and emotionally charged statements appear within the positive, negative and neutral review classes, which can answer questions like: what product qualities are most often touted, which are most often criticised and what about the customer experience elicit the greatest emotional response. The thematic aspect of the analysis provides value to the business intelligence that transcends the dichotomous nature of the question of a positive or a negative review.

9. To appreciate the difficulties of creating NLP systems with Java.

A key educational outcome of this project is to achieve a comprehensive and sincere grasp of the practical issues and constraints of developing NLP systems in Java as opposed to the Python ecosystem that has dominated modern data science practice. Although Java has much to recommend in terms of performance, scalability, security, and enterprise integration, its ecosystem of NLP library support remains less well-developed than Python's, with some features, such as access to the latest transformer-based language models, not available by default. By struggling with these limitations, finding real-world workarounds, and recording the trade-offs made would be an excellent addition to the educational goals of the project and reference information in other developers looking at Java-based NLP systems.

10. To propose possible enhancements and future applications of the model

The ultimate goal is to apply the lessons experienced during the implementation process, evaluation, and thematic analysis to come up with a tangible set of guidelines of how the sentiment analysis system can be enhanced and applied in the future. These recommendations involve expanding the sentiment lexicon to include the domain-specific vocabulary of musical instruments more exhaustively, integration of machine learning classifiers to supplement the rule-based system and make the system more accurate when dealing with ambiguous reviews, multi-language support to extend the scope of the system to non-English review markets, and aspect-level sentiment analysis which extends beyond document-level polarity to categorize sentiment towards specific product features on a case-by-case basis. The clarity of expression of these directions places the project in a wider research and

development path and reflects insight into the field beyond the direct scope of the work done.

1.3 Scope of the Work

The scope of this project has been established with a lot of care to make sure that the work is focused, can be done within the available time and resources and is also academically coherent. The ten items below are used to describe in detail what is contained within the project boundaries as well as what has been specifically omitted.

- **Focused Analysis on Musical Instruments Reviews**

The project itself is narrowly focused on the consideration of customer reviews attracted to the musical instruments category on Amazon. This controlled restriction of the analysis space fulfills several functions: it can be customized to the vocabulary and linguistic structure of this type of product; it can be more thematically focused and coherent in its analysis of customer opinion patterns; it does not produce its findings by washing out in a more general attempt to cover multiple product domains with different vocabularies, user demographics, and sentiment conventions. The project can produce more precise and practically helpful insights than a more diffuse approach would allow by focusing on a single, well-defined product line.

- **Scraped or Publicly Available Data were used.**

The information utilized in this project is limited to free datasets available in the open repositories or gathered on Amazon product pages in a morally and legally acceptable way. No proprietary, confidential or commercially licensed data is seen or used during any point of project. This limitation is both to ensure that the work adheres to relevant data usage policies, platform terms of service, and that it can be entirely reproduced by other researchers provided that they have access to the same sources of public data. The initial source of data and the most suitable to meet all of these criteria as well as offer a large enough and representative sample of authentic customer reviews is the Amazon Musical Instruments Review Dataset that is publicly available and has been widely used in scholarly research on sentiment analysis in the context of customer feedback.

- **Java use in NLP Tasks.**

Tasks This project is written in Java, unlike most sentiment analysis projects both in academia and industry, which are written in Python with frameworks like NLTK, spaCy, or HuggingFace Transformers. Being both a conscious technical challenge and a significant

educational contribution, this scope decision involves using a different and less well-documented set of NLP tools, mostly Stanford CoreNLP and Apache OpenNLP, and finding solutions to the more easily-addressed problems in Python. The Java implementation proves that text processing and sentiment classification with high quality are possible in this language and reflects that Java remains relevant today in the applications that merge the needs of enterprise software usage with the data analytics functionality.

- **Simple Sentiment Classification: Positive, Negative and Neutral**

The classification model which has been developed in this project is meant to categorize each review into one of the three sentiment classes which include positive, negative, or neutral. Although it represents a simplification of the entire emotional spectrum that customers can embody, this three-class scheme is well-established in the sentiment analysis literature as an effective and practically useful framework of converting unstructured opinion text into actionable business intelligence in a fairly interpretable way. Finer-grained emotion classification - the ability to differentiate between, say, excitement, satisfaction, disappointment, or anger is said to be out of scope in the current project, but is noted as a future direction in the recommendations chapter.

- **Rule-Based or Lexicon-Based Sentiment Analysis.**

The main method of analysis used in this project is not the supervised machine learning or deep learning but the rule-based and lexicon-driven one. This scope decision is based on the educational and transparency-driven goals of the project: lexicon based systems are more easily interpretable, easily debuggable and explainable, and do not need large labelled training datasets to operate. This sentiment lexicon is built based on pre-existing publicly available wordlists like AFINN, with additional domain-specific vocabulary to capture the vocabulary of musical instrument reviews. Although this method has had its limitations (such as decreased resistance to sarcasm, negation and out of vocabulary words), these limitations are clearly stated and met in the evaluation and recommendation sections of the report.

- **No Live or Real Time deployment.**

The present scope of the project is not that of the implementation of the sentiment analysis system as a live and publicly accessible service or interactive user-facing application. The system is a desktop-based scholarly prototype which parses a fixed dataset file and generates classification results and visualisations locally. The natural extension of the work, which would involve the creation of a web server backend, an API layer, authentication systems, and data ingestion pipelines, which must be running in real-time, is beyond the scope of what

could be accomplished with the time and resource limitations of this project. This demarcation is well recognized to ensure that the developers in the future who may have the desire to expand the system into a production context are informed on what more they would need to add to the system.

- **Visualisation Through Basic Graphs and Charts**

The sentiment classification results are represented graphically by a series of simple graphical outputs - pie charts, bar graphs, and word clouds - that describe the most significant trends in the data in a form easily understandable and interpretable. The project visualisation feature takes advantage of the inbuilt JavaFX charting functionalities instead of the third-party visualisation tools like Tableau or Power BI. Although this method generates easily understandable and communicative deliverables which can be used in the academic goal of the project, it lacks dynamic dashboards, filtering interaction, real-time data updates, and connection to external business intelligence systems. These higher levels of visualisation are found as possible improvements in future work discussion.

- **English Language Review Only.**

The reviews are analyzed only on the English language customer reviews. Multi-language support, including a preprocessing step to detect the language in input, independent sentiment lexicons in each target language, and possibly different tokenisation policies, are seen as outside the scope of the present project to ensure the complexity of the implemented NLP and sentiment classification processes remain manageable and the project is kept with a clear focus on the actual NLP and sentiment classification goals.

- **Single-Platform Focus: Amazon**

The project analyses reviews sourced exclusively from Amazon and does not extend to the aggregation or comparison of reviews from other e-commerce platforms such as Flipkart, eBay, Walmart, or specialist music retailer websites. This single-platform focus ensures consistency in the data structure, review formatting conventions, rating scale, and labelling scheme used throughout the project, and avoids the additional complexity of harmonising data from sources with different field definitions and quality characteristics. It also ensures that any platform-specific linguistic or behavioural patterns in the review data are handled consistently rather than obscured by cross-platform variation.

- **Academic and Research-Oriented Application**

The primary purpose of this project is educational and academic rather than commercial. The sentiment analysis system developed is designed to demonstrate the practical application of

Java in NLP, to develop the student's competencies in data collection, text preprocessing, classification implementation, and result interpretation, and to produce a well-documented academic report that contributes to the body of applied NLP literature. While the system could in principle be extended for commercial deployment with appropriate enhancements — including machine learning integration, real-time processing, and a web-facing user interface — the current version is intentionally scoped to the academic context and evaluated against the standards of research transparency, methodological rigour, and educational value rather than commercial performance.

1.4 Importance and Applicability

The sentiment analysis system created in this project has a meaning and possible applicability in a wide variety of stakeholder groups and contexts of use. The ten dimensions outlined below explain why this work is important both in practical, technical, educational and research aspects.

1. Assists Businesses to be aware of Customer Feeling.

1. Helps Businesses Understand Customer Emotions

Sentiment analysis at its simplest form offers an automated and scalable way of understanding customer perceptions towards their products and services, and brand experience in general. In the case of the manufacturers and sellers of musical instruments, this is because the capability to automatically categorize thousands of customer reviews by sentiment without investing time and cost in reading them one at a time gives them a continuous and complete view of customer satisfaction, which could not have been achieved under manual monitoring. The ability to detect dissatisfaction clusters around certain product characteristics, frequent compliments on certain tools, or new issues with product quality or delivery process provides businesses with the necessary intelligence to act in advance and make decisions based on evidence regarding product development and customer service priorities.

2. Improves Decision-Making for Product Development

The outputs of thematic and sentiment analysis in this project give manufacturers and product development teams a data-driven basis on what to prioritise to improve and innovate. Where the analysis indicates, such as that a disproportion of bad reviews of a certain model of keyboard mention tuning instability or key action weight, this is actionable intelligence, which can directly be utilized in future product development. Likewise, a positive sentiment regarding the sound quality of a certain line of guitars at all times

indicates that the given feature is a real competitive advantage that needs to be maintained and nurtured.

3. Favors Marketing and Branding Strategies.

The trends in the positive sentiment that the analysis reveals can be directly capitalized upon in marketing messages - determining what product features and user experience customers most commonly compliment can give a marketer an authentic and trustworthy set of messages that will interest potential buyers. On the other hand, negative sentiment patterns indicate the aspects of the customer experience that are most in need of redressing before they adversely affect the brand reputation on a larger scale. Sentiment tracking over time is also useful in allowing marketing teams to identify early warning signals of reputational risk during the post-product launch or service alteration, and to quantify the sentimental impact of individual marketing campaigns. The sentiment monitoring intelligence is a truly strategic asset in a competitive market place where brand perception is being increasingly influenced by peer review, as opposed to advertising.

4. Assists in Competitor Benchmarking

The analytical framework that is shown by the current project is directly applicable to competitive benchmarking applications, although the current project concentrates only on the in-house Musical Instruments review data of Amazon. Using the same sentiment analysis pipeline on reviews of competitor products posted on the same or similar platforms, businesses could build a methodical insight into the comparison of customer perception of their offerings with the offerings of their competitors on particular product features. This comparative intelligence - recognition of dimensions where competitors are highly positively reviewed and where there is a prevalence of criticisms - offers an orderly foundation on which strategic positioning choices are made as well as prioritising product or service improvisations that enable competitive weakness areas.

5. Improves online shopping experience

In the eyes of the final customer, systematic sentiment analysis of review corpora has the potential to enhance the online shopping experience. Instead of leaving individual buyers to peruse through hundreds of individual reviews so as to come up with their own evaluation of a product, an appropriately designed sentiment analysis system can reveal the most significant themes and the overall sentiment distribution in the entire collection of reviews in a succinct and organized overview. Review data are already a key input signal in the own recommendation and product ranking algorithms at Amazon, and more sophisticated

sentiment-aware recommender systems, which do not just match product recommendation to past purchase behaviour but also to the individual product properties that a buyer has previously positively responded to, are a natural extension of this functionality that the current project helps to illustrate and inform.

6. Demonstrates Practical Use of Java in Data Science

This project does have some unique contributions to the overall literature in sentiment analysis, but one of them is the fact that Java, which is not often seen as a language suitable to data science-related tasks, can also be successfully utilized to create NLP-related analytics systems. Most of the published sentiment analysis projects are written in Python, and the prevalence of Python in sentiment analysis leads developers with a Java background to think that data science applications are simply beyond their current skill base. This project broadens the perceived range of applications of Java, by showing that its fundamental capabilities of performance, scalability, robust object-oriented design, and an established enterprise integration ecosystem can be integrated with suitable NLP libraries to offer meaningful sentiment classification results to the Java-based application ecosystem.

7. Teaches About Natural Language Processing Concepts.

To students and developers working on this project, the practical nature of a sentiment analysis system gives them a practical, real-world exposure to the underlying principles of Natural Language Processing that are becoming the focus of computer science and software engineering. The preprocessing pipeline provides tokenisation, stop word elimination, and lemmatisation; the classification phase interacts with polarity scoring, lexicon construction and sentiment aggregation and the evaluation phase strengthens the comprehension of precision, recall, F1-score, and confusion matrix interpretation.

8. Applies in E-Commerce Customer Feedback Management

The outputs of this project can be directly applied to customer feedback management operations of the e-commerce firms. Receiving a significant flow of incoming reviews and support requests, customer service teams are left with a large workload, which can be meaningfully handled with automated sentiment classification. An analogous system to the one developed in this project is capable of minimising the number of manual labour required to manage feedback and also increase the responsiveness and uniformity with which the organisation can respond to customer dissatisfaction by automatically adding a sentiment label to incoming feedback, sending outrageously negative feedback to priority human

review queues and producing periodic reports summarising the trends in sentiment distribution.

9. Research in the field of Text Mining and AI

The papers and results of this project will be added to the existing literature of applied research in the field of sentiment analysis, text mining, and Java-based NLP. The project provides a well-documented case study of the design, implementation, and evaluation of a Java-based sentiment analysis system in one domain of specific and under-investigated products (the reviews of musical instruments offered by Amazon) and thus can serve as a reference resource that other scholars and software developers may rely on when working in the same field. The discussion of Java-specific implementation issues and the comparative analysis of various lexicon-based methods is explicit as well, which adds another aspect to a literature that is comparatively sparsely represented, due to the overwhelming prevalence of Python in published work in the field of NLP. The project is therefore a source of education as well as a contribution to methodology.

10. Scalable for Real-World Integration

Despite the present system version being implemented as a prototype designed and scoped as an academic project, the underlying architecture is deliberately structured as modular and extensible, so that it can be scaled to real-world application with proper further development. This separation of concerns between the data ingestion, preprocessing, classification and visualisation modules allows individual components to be upgraded or replaced, e.g. by replacing the lexicon-based classifier with a machine learning model, or by linking the data ingestion module to a live API feed, without necessarily re-architecting the entire system. This scalability of architecture, along with the existing enterprise integration and deployment capabilities of Java, makes the project a believable baseline to a production-grade sentiment monitoring tool instead of a demonstration project.

1.5 Role and Profile of Student

The student in this project played a broad role as the only developer and researcher of the project covering a wide scope of responsibilities in technical implementation, analytical reasoning, documentation, and professional development. The ten points below outline the particular contributions and competencies that were acquired in the course of this work. Knowing the Concept of Sentiment Analysis.

1. Understanding the Concept of Sentiment Analysis

The first step of the project was to come up with a comprehensive conceptual knowledge of sentiment analysis as a field of study what it is, its mechanics, what techniques are available and what can be done practically with it in a business and research setting. This knowledge was obtained by means of systematic consideration of the academic literature, technical documentation, and published tutorials, all of which allowed obtaining the theoretical background to make well-informed decisions regarding methodology, choice of tools, and evaluation criteria. Considering this conceptual base seriously before taking action on implementation made sure that the technical work was guided with a sense of purpose and not embarking on a trial and error approach.

2. Collecting and Preparing the Dataset

This student was to locate, source and compile the Amazon Musical Instruments review data utilized in the project. This entailed searching the existing public data repositories, comparing the candidate datasets with the project requirements based on size, completeness of the fields, quality of documentation and relevancy to the domain, and choosing the most suitable one. After the selection, the data set was downloaded and inserted into the Java development environment and the student performed an initial exploratory analysis to describe the structure of the data and define the quality problem and then planned the preprocessing steps that must be taken before starting the classification. This practical data preparation project helped to acquire skills in data sourcing, evaluation, and management.

3. Learning and Applying Java NLP Libraries

Much of the technical work of the project consisted in the research, study, and use of Java-based libraries of NLP — mostly Stanford CoreNLP and Apache OpenNLP — which were not before tools used in the programming practice of the student. This necessitated self-research of library documentation, trial with library APIs to learn what they can and cannot do, and the coordination of various library elements into a consistent and working processing pipeline. The practical experience of the Java-based NLP development, such as dependency management, API complexity, and lack of the comprehensive tutorials the student has access to with Python counterparts, significantly enhanced his/her knowledge of the NLP as an area of technical focus and Java as a language in a non-enterprise application environment.

4. Developing Logic for Sentiment Classification

The student developed and executed the fundamental sentiment classification logic that is the analytical heart of the project. It entailed building and filling the sentiment lexicon with

polarity-scored words that were suitable to the musical instruments review domain, writing Java code to tokenise and normalise every review, look up each token in the lexicon, sum up the resultant polarity scores, and apply threshold rules to give a final sentiment class label. The execution process was to be very sensitive to edge cases such as negation constructions, degree modifiers, and reviews with mixed sentiment among various product attributes. Creating this logic by hand, as opposed to using an existing machine learning model, gave a profound and hands-on insight into how sentiment analysis using lexicon works.

5. Designing Data Flow and Project Structure

In addition to coding individual components of the code, the student was to design the overall structure and data flow of the sentiment analysis system. This included breaking down the system into logical modules - data ingestion, text preprocessing, lexicon-based classification and result visualisation - and defining how the interfaces between the modules will be cleanly separated to maintain clean separation of concerns and maintainability. The modular design also makes the system easy to extend in the future since individual modules can be extended or substituted without interfering with the entire pipeline. The architectural structure of this plan and execution trained capabilities in software design that supplement the more NLP-specific competencies gained in other parts of the project.

6. Analysing Results and Visualising Outcomes

After running the sentiment classification pipeline on the entire review dataset, the student made a systematic analysis of the results and generated a collection of visualisations that conveyed the main results. This included the interpretation of the distribution of positive, negative, and neutral classifications on the corpus, the correlation between star ratings and the assignment of sentiment classes, and the patterns in the vocabulary most strongly related to each sentiment class by looking at word frequency data. The bar charts, pie charts and word clouds created in JavaFX were created to present these discoveries in an easily comprehensible form that could be presented to the stakeholders with no technical background, and this work developed the skills of the student in data storytelling and visualization.

7. Documenting Every Stage of the Project

All the stages of the project were documented in detail and continuously (selection of the data and preprocessing decisions to the implementation choices and evaluation outcomes) throughout the project life cycle. This documentation was a working record to the student through the development process and a basis of the final project report. Clearly and accurate

writing about technical decision-making, their justification, and the results that the student has observed trained the student in the capability to convey technical work to a professional audience, a skill that is of great importance in both academic and industrial settings. The documentation also guarantees complete reproducibility of the project by any researcher given access to both the same data source and development environment.

8. Troubleshooting Technical and Logical Errors

The implementation process invariably entailed coming across and addressing a variety of technical bugs, logical holes, and unforeseen behaviours in both the Java NLP library integrations and the bespoke classification logic. These problems could only be debugged by applying the systematic problem-solving skills and patience to track down where the unexpected classification outputs originated in certain preprocessing decisions, debugging Java dependency conflicts, and finding cases where the sentiment lexicon had wrongly or misleadingly scored domain-specific words. Overcoming these obstacles without solutions at hand, significantly enhanced the practical programming competence of the student, as well as created a more systematic and evidence-based way of debugging, which will be of great value in future professional software development efforts.

9. Learning Time Management and Self-Discipline

This work being a project carried out with a high level of autonomy over a long duration of time necessitated the student to have and practice substantial levels of self-management. Organising the project schedule, distributing the effort fairly among the various stages of work, when a specific technical methodology was not bearing fruit and a new one had to be considered, and being on time with documentation and report writing all required a degree of self-discipline and time management that is not part of the demands of more formal coursework tasks. These professional self-management skills, created under the immediate need to fulfill an independent project will be directly relevant to the requirements of professional work in the field of software development and data science.

10. Contributing to Personal and Professional Growth

Overall, the experience in designing, implementing, evaluating, and documenting this sentiment analysis project has contributed significantly to the personal and professional growth of the student. The project has increased technical skills in Java programming, natural language processing, data pre-processing, and analytical appraisal; conceptual knowledge in machine learning, text mining, and data analytics in e-commerce; and the softer skills of communication, documentation, problem-solving, and independent research.

Above all, the overall experience of working on a project based on a problem statement all the way to a working, evaluated and fully documented system has provided the student with a concrete and comprehensive reference point to the end-to-end practice of applied data science work that will inform and support future professional activities in the field.

<CHAPTER 2: REVIEW OF LITERATURE>

2.1 Introduction to the Review

The issue of obtaining and analyzing subjective human opinion in text has been the focus of long-standing research in computer science, linguistics, information systems, and marketing research more than 20 years. Sentiment analysis literature has evolved over the years since the primitive dictionary-based polarity scores systems to the present generation of transformer based deep learning models able to produce subtle contextual meanings. This chapter presents a systematic review of that literature, providing the development of the main methodologies, analyzing the results of major empirical studies and outlining the common themes and unresolved controversies that are of the greatest immediate interest in the methodology chosen in this project. The review is based on peer-reviewed journal articles, conference proceedings, and technical reports published in recognised sources such as Expert Systems with Applications, IEEE Access, Procedia Computer Science, the ACM Digital Library, and Springer Nature publications, and it focuses on publications dated 2015-24. Since this project is being implemented with Java, special consideration is also put towards Java-based NLP libraries and frameworks and the evidence that is available about their performance and applicability in sentiment analysis situations.

2.2 Concept of Sentiment Analysis

At the document level, the task is to label a single polarity, positive, negative, and neutral, to an entire review or text, based on the overall sentiment that the author expresses. The sentence level is an evaluation of each sentence in a document that allows detection of mixed sentiment between various sections of a text. Sentiment, the most informationally rich and granular form of analysis, is categorized at the level of particular attributes or qualities of the reviewed object of interest, e.g. in the case of musical instrument reviews, this could be the expressed sentiment about the sound quality, the construction of the build, the value for money, separately assessed in the same review text.

Rule-based systems are based on manually created sentiment dictionaries which assign polarity values to words or phrases, and a set of linguistic rules to deal with negation, degree modification, and other contextual considerations. Machine learning methods view sentiment classification as a supervised learning task, and train statistical classifiers on labeled training

set to learn the distribution patterns of each sentiment category. Deep learning models are neural network based models, such as recurrent networks, convolutional networks and transformer-based networks, which learn rich contextual representations of text, and have demonstrated state-of-the-art results on most standard sentiment datasets. The current project works mostly in the rule-based and lexicon-driven tradition, keeping in mind the machine learning comparisons and deep learning as the path to the future improvement.

2.3 Historical Development of Sentiment Analysis

The evolution of sentiment analysis over time. A scholarly gathering of support around the idea of computing sentiment in the early 2000s, for example, and when websites like Expe.net, social media, or user-generated media emerged with their growing numbers of online reviews, social interactions and the opportunities to continue, the data and motivation were ready to make for a fruitful research line. One of the earliest and most influential works, a 2002 paper by Pang, Lee and Vaithyanathan demonstrated that traditional machine learning classification algorithms, like Maximum Entropy, Naive Bayes and Support Vector Machines, could be applied successfully with the task of sentiment polarity classification in movie reviews, and effectively outperform a human-based baseline. This paper made the supervised learning paradigm a viable and effective way of sentiment classification and ignited the decade of active research in the machine learning approach to sentiment classification techniques. The other constructive modification in the very early history of the field was the construction of lexical resources specially devised to help out the rule-based sentiment analysis. Many of the initial lexicon-based systems were derived from the resources basic websites: SentiWordNet, an extension of the WordNet lexical database that adds sentiment polarity and subjectivity scores to individual synsets, and AFINN word list, numbers assigned to a selected set of English words that have emotional meanings. It was this notion that supportable formal, human labelled sentiment knowledge of the word level could be gathered and reused for applications and contexts in which any of the labelled training knowledge did not exist. The emergence of many social media sites and the amount of text-based content in the short form was exponential, and, in a short time, sentiment analysis moved from research and development to being marketed to some of the biggest businesses, powering brand tracking dashboards, customer care robots, finance market sentiment indicators and political opinion analysis systems. They started being applied in the field around 2013 and have revolutionised

the field of deep learning. More surprisingly, dense low dimensional vector representations of word similarities in the sense of semantic relationships were demonstrated to be able to considerably improve the performance of downstream natural language processing tasks like sentiment classification: this is what was proposed in Mikolov et al.'s famous Word2Vec model. In 2018, recurrent neural networks, attention and finally the transformer model, as introduced by Devlin et al. at Google, created a new state of the art further developed. Hailing from a long history of lexicon-related and classical approaches to machine learning, they are both practically available methodologies, interpretable and suitable for learning goals of the project and aim, being implemented via java environment.

2.4 Techniques in Sentiment Analysis

The rule-based sentiment analysis, the main methodology used in this project, builds the classification system out of a mix of a sentiment lexicon and a set of linguistic rules governing how the scores of individual words are combined and contextually modulated to make a document-level decision of whether a document is polar or not. The lexicon normally attaches a numerical measure of polarity to each emotive valenced word - positive values to words which convey approval, satisfaction, or enthusiasm and negative values to words which convey criticism, frustration, or disappointment and the general mood of a review is calculated by adding up the values of all the lexicon-matched words in the text being processed. The language rules can be easily interpreted, easily explained, are easy to customize to appropriate domains by extending the lexicon with domain specific words, and do not require labelled training data, which makes them pleasant to work with. The qualities of these features are congruent with the objectives of the current project for learning and transparency, and they align with the possibility of implementing the project in Java. The main weaknesses of rule-based methods such as lower resistance to sarcasm, inability to cope with domain-specific polarity reversal, sensitivity to out-of-vocabulary words are explicitly recognised in this project evaluation and forms the rationale behind the future improvements directions in the recommendations chapter. Machine learning-based sentiment classification methods learn statistical classifiers on labeled data, and learn the relationships between text features (usually word frequencies or TF-IDF scores) and sentiment classifications. In the previous sentiment analysis literature, several algorithms are tested and widely used including Naive Bayes, Logistic Regression, and the Support Vector Machines, and the latter two have been found to work best. Deep learning models, such as LSTM networks and transformer-based models, such as BERT, are the state of the art and mentioned in the literature review of this project,

which aims to put the rule-based approach into perspective and identify future opportunities to improve the model, but its computational demands and lack of support in the Java library are beyond the scope of the current implementation.

2.5 Java-Based NLP Libraries

While it is not the most widely used language in today's NLP/data science, Java provides several well-known, powerful libraries that allow for developing functional NLP/sentiment analysis applications. Of the three, Stanford CoreNLP, Apache OpenNLP, and LingPipe are the most useful, and most widely used, libraries that could be relevant for the scope of the present project, and have specific features and limitations that influenced the decisions made as to which tools were incorporated for the purposes of this work. Stanford CoreNLP is the oldest and most widely-used Java NLP library still usable, and is maintained and developed by the Stanford NLP Group. It offers a standard pipeline structure that can be used to perform a large variety of text processing operations such as tokenisation, sentence splitting, part-of-speech tagging, lemmatisation, named entity recognition, dependency parsing, coreference resolution and sentiment analysis with a pre-trained deep neural network. There is also the sentiment model to be used in CoreNLP, which was trained on Stanford Sentiment Treebank, which has the ability of giving sentiment scores to sentences and documents, and which can be used without additional training on the review text. The fine-grained features and the addition of detailed and informative documentation and community support make the library the most capable option for sentiment analysis from java, however, it has higher computations as compared to any simple choice and dependency management is prone to change when configured. Apache OpenNLP is an Apache Software Foundation Project; it is a lighter and more modular suite of NLP software focusing on the main NLP text processing actions for detecting sentences, tokenising, part-of-speech tagging, chunking and recognizing named entities. Although sentiment analysis is not built into OpenNLP as a pipeline element, the classification framework of the library can be trained on a label-based sentiment classifiers and the tool can be a useful solution in a machine learning driven sentiment analysis in a Java program. The third alternative, LingPipe, is created to handle large amounts of text and offers classification, named entity recognition and topic modelling functionality as customisable pipeline elements. The combination of Stanford CoreNLP due to its ready-made sentiment and linguistic analysis features and a set of Java logic due to the lexicon-based scoring is the most

efficient one that can be offered in the context of the given project and has therefore been chosen as the major implementation strategy.

2.6 Sentiment Analysis in E-Commerce

The application of sentiment analysis to e-commerce data of customer reviews has been one of the most useful and commercially promising applications of the applied NLP literature, and it has inspired both methodological innovation as well as commercially worthwhile insights about the link between stated opinion of the consumer and commercial outcome. The original article on feature-based opinion summarisation by Hu and Liu (2004) gave a highly influential model of aspect-level analysis of product reviews, revealing that this customer opinion could be broken down into opinions on product aspects, and more informative than just document-level polarity. Stabekk's approach is based on the aspect level and has been developed and refined from various other studies and is one of the most active research strands in the field. Archak and colleagues analysed the quantitative correlation between the content of Amazon product reviews and measured commercial consequences, and the general valence, i.e. the valence of the sentiment conveyed in the single reviews, as well as the aspect-based breakdown were shown to be predictive of the product sales figures. Not only descriptive, but also having actual predictive value for business decisions, with the implications of this finding significant to the current project. More recent studies since 2020 have concentrated on the incorporation of sentiment into a personalised recommendation system where the sentiment, specifically towards different product attributes, of individual users is used in order to make personalised recommendations that are more attuned to the actual preferences of the user than just past purchasing activity - a path towards one of the most economically promising possible applications of the sentiment analysis framework developed in this project. Literatures focusing on the reviews of musical instruments and/or other creative product literatures is rather rare in the published literature and is more likely to be focused upon areas of extremely high numbers of reviews, such as electronics, restaurants and movies. This is a byproduct of the literature review since it sets the arena in which this project is making a unique and important contribution, specifically in a relative absence of domain specific work. As mentioned, whereas the analytical problems presented in that chapter, and made difficult by the requirement that musical instrument reviews offer emotional expressions and demonstrate technical expertise, and contain both positive and negative sentiment elements, were not a part of the previous e-commerce sentiment analysis literature, the present work fills a gap in

expanding the knowledge base of sentiment analysis by demonstrating what is possible with a Java-based lexicon and rule-based tools in this area.

2.7 Challenges Highlighted in Literature

The challenges that are most consistently recognized in the literature and that would correspond well with the specific challenges of sentiment analysis for Amazon Musical Instruments would parallel those problems encountered with sentiment analysis in general. Ambiguity in language, i.e. the fact that one and the same word or phrase can have different implications of sentiment based on the context in which it is used is generally recognized as one of the most basic barriers to the successful sentiment classification based on rules. This issue is especially acute in the field of musical instruments due to the dual sense that a number of words might have: the term light is positive when applied to state how heavy a travel guitar is but negative when applied to say how loud an amplifier is; the term sharp is positive when applied to the accuracy of intonation and negative when applied to the feel of the fret ends; the word bright can be used to characterize a desired quality in the sound of Ambivalent feelings in individual reviews is a cross issue that the literature has always found to be detrimental to the accuracy of document-level classification. It's the ones that are neither too positive nor too negative and lend the prospective buyer a more balanced view that are the most informative and useful ones. These reviews don't straightforwardly fit into the label of either a positive or a negative rating on the document level and labelling a single polarity on the entire document will always bias some part of the review content. But in principle, most efficiently achieved with "aspect-level analysis" such as those followed by Hu and Liu and their successors, this limitation can be (indeed has been) circumvented by requiring a more complex system of analysis than the document-level classification system employed by the present project. In this project, the domain dependency is also a problem that comes from lexicon-based algorithm, "Sentiment Classifiers trained and calibrated on one kind of text are not well suited for a different domain. General purpose sentiment lexicons such as AFINN are constructed from a variety of text types and may not have an appropriate score for the domain-specific items as the sentiment value in the musical instruments domain is not the same as general language. The systematic use of words specific to the selected domain and the difficulty resilience of words which have clearly ambiguous valence in this domain is an interesting factor to consider for mitigation, which is incorporated into the lexicon building method of the project. It was one of the most challenging task in the field, and the current development of the sarcasm

detection system is consistent with the existing literature in the field, which shows that a powerful sarcasm detection system using a rule-based approach or classic machine learning approach is not so far a reliable real-world possibility.

2.8 Comparative Analysis of Tools and Techniques

The existing sentiment analysis tools and methods being used or evaluated – such as rule-based systems, machine learning classifiers, and deep learning architectures – provide insightful background in order to be able to understand the methodological decision taken in the current project and the trade-offs the decision entails. The next table recaps the major dimensions these approaches vary based on using the empirical results of the literature examined in this chapter.

Technique	Accuracy	Flexibility	Java Support	Complexity
Rule-Based / Lexicon	Medium	High	Yes	Low
Machine Learning	High	Medium	Limited	Medium
Deep Learning (BERT)	Very High	Low	No (mostly)	High

Table 2.1: Comparative Analysis of Tools and Techniques

As the table indicates, lexicon driven/rule based is the most adequate major methodology that must be applied in an academic project of this kind developed based on java programming language. While deep learning models such as BERT, when fine-tuned with large, labeled datasets, yield the highest level of accuracy, these are implemented in Python and are computationally intensive, not easily understood, and don't fit well into the design of a project to display the NLP power of the Java language and its outputs that are dialog-driven and simple to understand and explain. Machine learning classifiers require less labelled data than Python or Java classifiers, have less Java library support, and are more accurate than the rule-based classifiers – but they are not as fast as the rule-based classifiers and are still a viable improvement. The rule-based is not at the top end of accuracy but has the most Java compatibility: it is easily interpretable and easily implementable based on the context of the project.

2.9 Research Gap Identified

A close examination of the literature on sentiment analysis that is currently available shows that there are a number of very important gaps that generally characterize the area within which this project will make its contributions. The gap which can be immediately utilized is the lack of published sentiment analysis work which is performed in java language. The widely used NLP libraries provide implementation of most of the NLP studies, and even among studies done in java only a few are discussed and mostly fall in category of enterprise information extraction problem and are related to consumer sentiment analysis in e-commerce environment. This also means that the developers and researchers who use Java frameworks to create NLP projects may have limited access to reference materials to refer to in implementing NLP techniques, and the published literature's ability to describe the strengths and weaknesses of the NLP library ecosystem for creating sentiment classification in Java is not well established. This gap will be directly addressed in this project, as it provides a case study on sentiment analysis using java language which is well documented, well evaluated and well-documented. Niche segments like musical instruments, where the number of reviews is less, vocabulary is more technical and the buyer segment carries unique features have been relatively under-researched. Both areas of analytical complexity—the use of the terms that reviews employ (valued as double-sided features) and the amount of mixed-sentiment information, as well as the expressiveness of the population of buyers—are not adequately accounted for in the literature, and the deliberate focus of the project on analyzing such reviews constitutes a specific and original addition. The third gap is related to the treatment of Java as an efficient and competent environment to perform data science applications. Most of the documentation about practical implementations of NLP and ML that is available, either assumes Python is used to implement the applications or doesn't mention what is actually used, and so the notion that the only meaningful thing to do is to do it in Python is reinforced. It offers concrete, practical demonstrations of what can be accomplished in Java using existing libraries and demonstrates workarounds and challenges that arise from using these existing libraries, while also showing scripts can help achieve the same performance as the existing libraries, and the performance achieved by using the library system and scripts can be compared to existing guidelines.

2.10 Conclusion

This chapter has provided a literature review that has traced the development of sentiment analysis since the very beginnings and through the era of machine learning to the current era of deep learning supremacy on which the current project is based. Considering the requirements of the academic project, the rule-based and lexicon-driven approach were identified as the best primary methodology to be applied in the project since it is easily interpretable but more difficult to implement, and it is also compatible with Java. The challenges which are unique to the analysis of this kind of data (sarcasm, mixed sentiment, domain-specific vocabulary, language ambiguity) have been recognised and traced to the general literature on sentiment analysis challenges. Each of the existing instruments and methods have been evaluated in a comparative way, which has enabled a theoretical account of the methodological decisions that were made during the project. The research gaps have helped in elucidating a particular contribution that they will make to current literature. The implementation choices presented in the following chapter have been assessed and discussed with regards to their relevance to the project needs, and are provided with technical support from the Java NLP libraries: Stanford CoreNLP and Apache OpenNLP. The insights about the available analytical possibilities, as well as the practical constraints that are implemented due to this review, place the project to continue with its empirical implementation with the correct expectations regarding what the system can and cannot do, as well as a clear roadmap of the possible improvements that would be most fruitful in extending the work in subsequent iterations.

CHAPTER 3. RESEARCH OBJECTIVES AND METHODOLOGY

3.1 Research Objectives

The aim of this project is to effectively and reliably classify the customer opinion expressed in the text of the reviews by designing, implementation and evaluation of a sentiment analysis system. The project particulars the class of evaluations on Amazon Musical Instruments that are categorized as reviews, and constitutes a combination of lexicon-primarily based and supervised machine studying strategies to achieve the polarity cues and convert raw textual content into data suitable for business intelligence. The project has been structured in terms of a series of distinct, quantifiable, and mutually-dependent goals, which jointly address the whole data science lifecycle between data acquisition and model deployment and insight communication. The first is to collect, scrub and preprocess a representative sample of the data from Customers' reviews. The first involves collecting, scrubbing and pre-processing a representative sample of the data from Customers' reviews that can be used to classify sentiment. This includes access to a publicly available dataset containing authentic buyer reviews, exploratory data analysis to learn aspects of distributions and quality issues, and a complete data pipeline for pre-processing the data, including noise removal, text normalization, tokenization, stop word filtering and lemmatization. Proper implementation of this goal will ensure that all subsequent modelling will be supported by high quality and well-prepared modelling inputs – rather than raw or poorly formatted and structured text. The second item, is to train several of the sentiment classification models on the ready dataset and this will incorporate lexicon-based models (textblob and vader) and also the classical supervised machine learning models (naivebayes, logistic regression, support vector machines and random forest). All of them are analytical traditions and it ranges from the rule-based polarity scoring to statistical pattern recognition. It is implemented in best practices with Python and standard library, with sentiment labels based on star ratings to facilitate supervised evaluation. This goal will make sure that the comparative analysis that will be made later will be based on well-configured and consistently implemented methods. The third goal is to compare and evaluate the predictive performance of the approaches, based on quantitative standardised metrics. It is evaluated on a held-out test partition that is not used in training the model to ensure the accuracy value is due to the model's generalisation capacity and not in-sample memorisation. All measures of success reported are the accuracy of classification, the precision, the recall and the F1-score and the confusion matrices provide a

more detailed analysis of the behaviour of the classification on all three classes of sentiment. It is this goal that provides the evidence base on which they base conclusions about the relative effectiveness of methods in the project. It includes analysis of distribution and polarity scores histograms, sentiment distribution plots, word-level analysis via word clouds and frequency analysis. The visualisations have two roles: interpretative (what drives sentiment in this product domain?); and diagnostic (what problems are there that differentiate processing/summary statistics from modelling?); and neither function could have been satisfied using only summary statistics. The fifth and last goal is to create a detailed report of the whole analytical procedure, generalise the results to the stakeholders in the e-commerce sphere and to determine the ways to further improve the methodology.

3.2 Research Problem

The major research challenge to solve with the project is how to interpret the sentiment in large volumes of unstructured text such as consumer reviews, in a systematic and scalable manner. Much of the traditional way of determining customer feedback, such as via reviews, asking for feedback through a survey or focus groups, is labor-intensive and often limited. While a single analyst can take in a few reviews in meaningful ways within a time horizon, the size of the data becomes hard to manually sentiment code. Meanwhile, the amount of such review material that has been published both in reviews on sites like the Amazon, as well as other means, continues to rise and keep increasing, which add to the disconnect between what information may be available and the organizations' capacity to use it in traditional manner. This disjunction inspires a data-driven strategy that is automated. Nevertheless, it is not an easy task to automatically classify natural language text sentiments. A variety of different forms, formalities, lengths of customer reviews and different forms, variation in sentiment information represented in customer reviews. Different polarity within a customer review (as in there is the use of negation, sarcasm, mixed polarity etc.) which may or may not be well-represented in general purpose sentiment data, and the use of domain specific terms that may or may not be well-represented in general purpose sentiment resources. The dataset which is researched in the case of the reviews of Amazon Musical Instruments, has an added complexity, because the level of feedback regarding the product which belongs to this category of musical instruments is very emotionally charged, and, on another note, technically detailed; one comment to an instrument can include both sides and be both positive and negative at the same time.

3.3 Research Design

The type of research design used in this work is a quantitative and empirical research design based on the comparative case study approach that is the norm in applied machine learning and NLP study. The design of the methodology is based on the controlled experiment where several sentiment classification methods are used on the same dataset with the same preprocessing and evaluation conditions allowing the systematic and unbiased comparison of the relative performance of the methods. The design is data-driven such that conclusions are drawn due to trends in model predictions and/or metrics of evaluation rather than purely through theoretical derivation. An example of a single-dataset design would be using the same dataset of Amazon Music Instruments reviews on which all the methods are trained and tested. All the phases are described in great detail to allow for reproducibility. The design is a mixture of exploratory and confirmative, using exploratory data analysis for describing the corpus and making decisions on the approaches to be used for preprocessing, and confirmative by learning how well supervised approaches to machine learning will perform on the labelled review data set, against that of lexicon approaches, and comparing results in line with the hypothesized improvement, which was detailed in Chapter 2.

3.4 Type of Data Used

This research project will involve secondary, quantitative data that will be cross-sectional data pertaining to customer reviews of product category: Musical instruments on Amazon. The data is secondary in the sense that it was created and recorded by the users of the platforms and the review infrastructure of Amazon itself and not created by the researchers themselves (primarily, with the help of surveys or structured interviews). The data is quantitative as it combines both numerical fields such as the star ratings on a one-to-five scale, the number of votes on the helpfulness of the review, and Unix timestamps with non-numeric or categorical or text-based fields, such as the review body, reviewer name, product identifier and the sentiment label assigned automatically due to the rating. Amazon Musical Instruments Review Dataset (AMIR) is a well-documented, publicly available corpus of reviews that have been used in various published sentiment analysis experiments, and also used in the work of this project. The length of the reviewed texts varies widely: the one-sentence shortings are varied with the multiple paragraph length, which are more informal, more technical, and focused on the products. This variation renders the corpus realistic and methodologically challenging in that preprocessing and feature extraction choices need to be resistant to the entire gamut of text styles in the data. The labels of sentiment are based on the

numerical ratings of stars that are given to each review. “Four or five star reviews will be positive sentiment (high satisfaction); three star reviews will be neutral (mixed or indifferent review); one or two star reviews take on a negative sentiment (dissatisfaction).” This mapping from rating to sentiment follows the trend of most published Amazon sentiment analysis studies, and provides a principled labeling protocol as well as an intuitive and reproducible labeling schema, without requiring to manually label the entire dataset.

3.5 Data Collection Method

The data collection methodology that will be used in this project is coupled with retrieval and utilization of an already available publicly accessible dataset as opposed to primary data collection whereby it will be collected by interacting with the subject of the review directly. The Amazon Musical Instruments Review Dataset comes from a large collection of Amazon product review data across various categories which are stored on the dataset repository maintained by Jianmo Ni at the University of California San Diego and has been widely used in various published NLP and sentiment analysis research papers. Clean and structured files/JSON, CSV files with a schema description in fields and data provenance are present in this repository. The data was downloaded in CSV format and imported into Python for analysis using the Panda library. The data has been fetched from a file in csv format and loaded into the python environment for analysis using the pandas library. The downloading process was straightforward; it required only a common web browser, a good Internet connection and the downloaded document provided a complete set of review records, complete with all the necessary fields. Prior to analysing the data, the documentation was examined in the form of a data dictionary to ensure the meaning of each field, any encoding conventions and any known problems with the data quality were understood by the person undertaking the analysis as stated by the dataset's curators. The number one benefit of this data that it has been proven and community-tested is the possibility of reproducibility: Head into the repository with the same repository URL, any researcher can obtain the same source data and apply the following procedures of analysis described in this report. Here lies this level of reproducing, one of the basic components of transparent research work, particularly of this project in which the part played by the methodology is the comparison of the various approaches to classifiability; the quality of the comparison then depends on common data base of all the methods.

3.6 Data Collection Instrument

The data collection methodology that will be used in this project is coupled with retrieval and utilization of an already available publicly accessible dataset as opposed to primary data collection whereby it will be collected by interacting with the subject of the review directly. The Amazon Musical Instruments Review Dataset is retrieved from the dataset repository maintained by Jianmo Ni at the University of California San Diego, which contains large amount of data on the product reviews posted on Amazon in various categories, and has been widely exploited in the published literature of NLP and sentiment analysis. It includes clean and structured JSON and CSV files that are documented to explain the schema in the fields, and the generation of the data. The data was downloaded in comma-separated values (csv) format and provided to the python data analysis environment using the Pandas package. The downloading process was simple and only needed a common web browser and a steady internet connection and the resultant file had a full set of review records with all the pertinent fields intact. Documentation for the data, provided in the form of a data dictionary, was administered before the analysis to ensure that the meaning of each field was known, coding rules were known and any broken quality items as recorded by the curators of the data set were considered. Printed or community-tested data like this provide one of the primary advantages of reproducibility: any researcher, with the same repository URL, is given access to the same source data and uses are allowed to replicate the analysis procedures described in this report. This is the level of reproducibility, that is an indispensable part of transparent research work and particularly of the project in which the contribution of the methodology is the comparison of the various orientations to classification; this comparison is only good if all the used methods rely on the same data base.

3.7 Sample Size

The size of the sample that can be used by this project will be the number of complete and usable review records in the Amazon Musical Instruments dataset following the preprocessing and filtering by quality. Overall, there are approximately 10 000 records of reviews, representing one customer transaction rated with a star rating, text review and metadata. After applying the processing operations (which may include elimination of records with blank review bodies, elimination of duplicate records, as well as any extreme outliers), the sample consistently processed can be modelled comfortably using the modelling methods employed in this project with the size ranging between 9,000 and 10,000

observations. The concept of sample adequacy is applied in the supervised machine learning classification context of the dimensionality of the feature space, there's a number of classes that needs to be predicted and computation needs of the selected algorithms. This dimension of the effective feature space can grow to thousands of dimensions after vocabulary construction, but by setting a minimum document frequency threshold and a maximum of features, dimensionality reduction is possible in this regard. The number of records is about 9,000 to 10,000 (split approximately 80/20 for training and testing) which is well in the range of those printed in other examples that have similar classification procedures and data sets. Given that the Amazon reviews were previously found to be positively skewed, the class distribution is not even in this sample. According to the preliminary analysis of the data, there are more positive reviews than neutral and negative reviews, with the number of 5-star ratings being the majority of the data body. The unbalance is addressed during modelling by employing class weighting in training and evaluation, while the adoption of the class weighting and reporting of the F1-scores measures in place of accuracy is carefully considered when choosing the evaluation measures to provide an honest evaluation of the classifiers performance in each of the three sentiment classes.

3.8 Sampling Technique

The sampling method used in this project is not sample set 1 (the whole of raw information) of the prepared data, but rather the split of the whole prepared data set (set 1) into training (70% to 90%) and testing (10% to 30%) sets for model development and testing. This partition is performed using "stratified random sampling" performed using the "train_test_split" function of the Scikit-learn library where the "stratify" parameter is given the sentiment label column. Stratified sampling will ensure that trainset and testset's positive, neutral and negative class representation matches that of the whole data set. This is particularly important because the review corpus from the Amazon is not homogeneously balanced in terms of the number of reviews in the different classes: If the reorganization of the review corpus was performed (simple random sampling), then the classes with few reviews (few negative and few neutral reviews) would have been highly probable underrepresented in the test set, leading to unreliable estimates of the performance of those classes, in the case that the performances of these classes were assessed. Also, the sentiment labelling of stratified sentences makes sure that the partition of both the training set and the test set are indeed representative of the entire distribution of class labels and that the results of the evaluation do not rely on artifacts of an unrepresentative partition. The split ratio used

is the standard ratio used in the classification literature, 80 percent training and 20 percent test, which provides adequate training, and sufficient test observations to assure statistically significant and effective performance estimates and consistency. Additionally, calling train test split assigns a deterministic random seed to make sure that the same split is generated every time the code is run (one of the prerequisites for the reproducibility of the results of a comparative evaluation). Such reproducibility will also be required for scientists to check the work of others and for scientists to find it easier to distinguish among the many different modelling runs undertaken during the hyperparameter configuration phase.

3.9 Data Analysis Tool

Main data analysis environment used in this project are Python Programming Language, as Python is becoming the standard platform for data science projects and NLP based apps with the extensive ecosystem of its specialised libraries provides balance of readability and computation power and facilitates interactive exploration and analytical processes. All data manipulation, statistical analysis is done in Python in the Jupyter notebook interactive development environment, all models are trained, assessed and visualized in Python. Here are the major libraries which make up the different stages of a project. Pandas offers the leading data loading, inspection, cleaning, transformation and aggregation functions, and works with DataFrames, which offer an efficient and adaptable interface to work with tabular review data. Most of the downstream steps involve numerical calculation and operation on arrays, which is supported by NumPy. Matplotlib / Seaborn are used to generate all the data visualisations, including distribution histograms, bar charts, pie charts, word clouds and the confusion matrix heatmaps. The text processing platform is provided by the Natural Language Toolkit (NLTK), and the spaCy library provides domain-dependent components for reviews such as English-language lemmatisers and stop word lists for reviewing text. All four of the supervised classifiers—Naive Bayes, Logistic Regression, Support Vector Machine, and Random Forest—along with TF-IDF vectorisation, train-test splitting, cross-validation, grid search of hyperparameters and the full suite of classification metrics like accuracy, precision, recall, F1-score and confusion matrices are implemented in the scikit-learn machine learning core of the project. The sentiment analysis system is obtained using VADER using Python package vaderSentiment and TextBlob has a sentiment scoring API. These tools create comprehensive open source analysis framework without dependency on proprietary software that can be reproduced entirely without any proprietary software.

3.10 Procedures of data preprocessing and feature engineering

Data preprocessing is the first step in the methodology since the quality and suitability of data supplied to the sentiment classification techniques are the main factors determining the validity of the results. The first step of the pipeline is to examine the raw data to understand its structure (variables and data-type) and quality issues. It will involve reviewing the top and bottom items in the DataFrame, examining data types, summarising any numerical columns and providing counts of missing values for all fields. At this point any record without Review text fields or containing blank reviews is dropped because it might not provide any meaningful information to the classification process. This involves taking out HTML tags and coded entities that can be found in reviews gathered via web interfaces, eliminating hyperlinks, eliminating special characters and punctuation, lowercasing all text, and eliminating numerical tokens that do not have sentiment data. Emoticons and emoji characters are either removed or communicate clearly their sentiment and when they are features, they are replaced by text tokens that mean what they mean. Emoticons and emoji characters are removed, or with clear sentiment signal are replaced with descriptive text tokens then deleted. Now the text is tokenised with the NLTK word tokeniser, tokenising the whole cleaned review into a list of word tokens for further processing. After tokenisation, a standard stop word list in English is used (enhanced with domain-specific high-frequency words that do not have any value of discriminative sentiment in the musical instruments context) to eliminate function words. NLTK WordNetLemmatizer Lexicist minimises the number of tokens by reducing to the base form the inflexible forms of a word and reduces the number of terms in the effective lexicon by eliminating the lexical difference between the similar words. The supervised machine learning classifiers are feature engineered using the following steps: Processed token lists to numerical feature vectors using TF-IDF vectorisation using the Tfidf Vectorizer of Scikit-learn. The vectoriser is specified with minimum document frequency to exclude extremely rare words as 3, feature limit to deal with dimensionality as 15000, unigrams and bigrams to capture Phrasing sentiment Expression (PSE) along with Single word indicators as ranges. For the lexicon-based models (VADER, TextBlob), we simply pass the cleaned but not vectorised text into the tools, as they accept the text as their input. The derived polarity compound score of VADER and sentiment polarity value of TextBlob are both thresholded to give positive, neutral or negative class labels in the format of supervised classifier results, allowing one to easily compare all method.

3.11 Framework of Model Training and Evaluation

The model training and evaluation system is enacted to give a principled, fair and reproducible foundation to the performance of all the sentiment classification methods. After preprocessing and feature engineering, training each supervised classifier is based on model fitting the training sub-set of the TF-IDF feature matrix with the sentiment labels. Each of the classifiers is optimized using a five-fold over-cross validated grid search with observations done with changes to the parameter grids for hyperparameter optimization. In the case of Naive Bayes, the smoothing parameter alpha is changed in a logarithmic grid. In the case of Logistic Regression, regularisation strength C and the maximum iteration limit are changed. The regularisation or C is adjusted in case of Support Vector Machine. For Random Forest, the number of trees and the value of the Max. depth and the Min. samples/leaf values are varied. In each of these, the optimal choice of the configuration over the cv estimates the F1-score, and the best cv configuration in the group is the configuration that is used for the comparative evaluation, giving headway to the optimal (not default) performance of each of the algorithms. In the lexicon-based methods, VADER is used on each review in the test set based on its compound polarity score: reviews with a compound score greater than 0.05 will be classified as positive, reviews with a compound score less than -0.05 as negative, reviews with a falling within the threshold range as neutral. Analogous thresholds are applied to TextBlob polarity attribute. Lexicon-based sentiment analysis is an unsupervised approach, trained on no data! The objective part is the only part which is assessed in the model. The overall accuracy, class-wise precision and recall, macro-average and weighted F1-score, and a three by three confusion matrix that breaks down classification performance per class versus true/predicted class combination are used to compare the results of both methods to the true sentiment labels. To support the numerical measures reported, and to provide a sense of “how you are doing” to the user, “confusion matrix” outputs are generated as visualisation measures in the form of heat maps. The histograms of the polarity scores of the lexicon-based methods and the feature importance of the highest scored TF-IDF terms for each model in the inspection of the highest-scored terms are other possible outputs.

3.12 Summary

This chapter has given a clear specification of the research objectives, problem statement and research methodology that will dictate the process of undertaking this sentiment analysis project. It contains five different objectives: (1) Data acquisition and preprocessing, (2) Multi-method sentiment classification, (3) Standardized assessment and evaluation, (4)

Exploratory and diagnose analysis, (5) Documenting with business-relevant recommendations. The research question outlined is: How can the sentiment classification problem be specified for Amazon customer reviews that can be classified at a large scale and with high reliability despite the problem's linguistic, data quality and method-selection aspects? The research question posed in this research is: How can this problem be formulated regarding sentiment classification of Amazon customer review text for reliable and scalable classification at very large scales that the challenge posed by problem aspects such as Linguistic problems, Data quality problems, and Method-selection problems are acknowledged? The study design is quantitative and comparative, which involves several classification methods to a single regularly preprocessed dataset controlled. The data is the secondary record including publicly available data of Amazon Musical Instruments reviews and sentiment labels according to star ratings. The working sample is a stratified random sample made up of approximately 9 to 10 thousand records, training to test ratio is 80:20, this sample is class-proportional and is reproducible by setting the value of the random seed. All of the toolbox tools are completely built using Python libraries such as Pandas, NLTK, Scikit-learn, Matplotlib, Seaborn, VADER, and TextBlob. The pre-processing steps involve noise removal, text normalisation, tokenisation, stop word removal, lemmatisation, TFIDF for supervised classifiers and direct text input for lexicon-based classifiers. The model training and model evaluation system is the cross validated hyper parameter optimization of all the supervised classifiers, the final analysis is done in the held out test set on the basis of accuracy, precision, recall, and f1 score along with confusion matrix analysis. This methodological underpinning is extensive and the project will thus have the potential for generating robust, meaningful and applied findings (these will be presented in subsequent empirical analysis chapter).

CHAPTER 4. DATA ANALYSIS, RESULTS, AND INTERPRETATION

4.1 Introduction

In this chapter, the emphasis is put on the practical implementation and analysis of the sentiment analysis project that was carried out on the customer reviews of musical instruments that are offered at Amazon. For the project, a main programming language used is Java, which was used to design a system that can determine if a customer's review is positive, negative or neutral. The system architecture, the tools and techniques used, the steps of implementation and a detailed analysis of the results are discussed in this chapter.

4.2 System Architecture and Tools

All the modules of the sentiment analysis are implemented as a Java application with the following major modules.

Data Collection and Preprocessing: This module processes the raw customer reviews dataset, and cleans up the data by cleaning and normalizing the text.

Feature Extraction: Transforms textual information into numerical features which can be used in machine learning, and employs methods such as Bag of Words and TF-IDF.

Model Training and Classification: Uses machine learning models, e.g. Naive Bayes or Support Vector Machines, to train the labeled data and make sentiment predictions.

Evaluation and Reporting: Measures the performance of the model and it provides valuable insights using the sentiment predictions.

The rich ecosystem of Java, such as libraries such as Apache OpenNLP, Stanford NLP, and Weka offer a good foundation to develop such a system

```
dataset.head(n = 10)
```

	reviewerID	asin	reviewerName	helpful	overall	unixReviewTime	reviewTime	reviews	sentiment
0	A2IBPI20UZIR0U	1384719342	cassandra tu "Yeah, well, that's just like, u...	[0, 0]	5.0	1393545600	02 28, 2014	not much write exactly supposed filter pop sou...	Positive
1	A14VAT5EAX3D9S	1384719342	Jake	[13, 14]	5.0	1363392000	03 16, 2013	product exactly quite affordable not realized ...	Positive
2	A19SEZSQDW3E21	1384719342	Rick Bennette "Rick Bennette"	[1, 1]	5.0	1377648000	08 28, 2013	primary job device block breath would otherwis...	Positive
3	A2C00NNG1ZQQG2	1384719342	RustyBill "Sunday Rocker"	[0, 0]	5.0	1392336000	02 14, 2014	nice windscreen protects mxl mic prevents pop ...	Positive
4	A94QU4C90B1AX	1384719342	SEAN MASLANKA	[0, 0]	5.0	1392940800	02 21, 2014	pop filter great look performs like studio fil...	Positive
5	A2A039TZMZHH9Y	B00004Y2UT	Bill Lewey "blewey"	[0, 0]	5.0	1356048000	12 21, 2012	good bought another one love heavy cord gold c...	Positive
6	A1UPZM995ZAH90	B00004Y2UT	Brian	[0, 0]	5.0	1390089600	01 19, 2014	used monster cable year good reason lifetime w...	Positive
7	AJNFQI3YR6XJ5	B00004Y2UT	Fender Guy "Rick"	[0, 0]	3.0	1353024000	11 16, 2012	use cable run output pedal chain input fender ...	Neutral
8	A3M1PLEYNDEYO8	B00004Y2UT	G. Thomas "Tom"	[0, 0]	5.0	1215302400	07 6, 2008	perfect epiphone sheraton ii monster cable wel...	Positive
9	AMNTZU1YQN1TH	B00004Y2UT	Kurt Robair	[0, 0]	5.0	1389139200	01 8, 2014	monster make best cable lifetime warranty does...	Positive

Fig 4.1- Dataset from Amazon's Musical Instruments category

Project Goals:

1. Preprocessing text: Transforming raw text reviews into structured formats that can be analyzed. Techniques included:
 - o Removing special characters, numbers and punctuations.
 - o Text to text rematives: Stop word removal, Lemmatization and Tokenization of text to standardize it.
2. Feature Extraction: Extract meaning representations of text data – using any of the following methods:
 - o Bag-of-Words (BoW)
 - o Term document frequency (TF).
 - o Document frequency (IDF).
 - o TextBlob scores of sentiment quantification by Polarity.

3. Model Building and Classification: Train different machine learning models for effective classification of customer sentiments.
4. Evaluation and interpretation: Assess the working of models with using accuracy, precision, recall, F1-score and confusion matrix.

Importance of the Study:

The project is real-world applied to the business particularly in the e-commerce platform. The feelings expressed in a passage can be automatically categorized:

Customer Insights: Companies can gain insights into customer satisfaction trends and decide where they need to make improvements.

Product Improvement: Specific improvements can be made to products and services, identify the most common issues when evaluating products/processes.

Marketing Strategies: Positive Sentiment Net Trends can be leveraged to create marketing campaigns and negative Sentiment Net Trends can be used to create damage-control campaigns.

Automation: Sentiment analysis models can be used to decrease the hard work of monitoring for manual review and save resources.

Challenges Addressed:

Textual Variability: Addressed problems associated with the diversity of stylistic devices used in writing, spelling errors and using abbreviations for customer reviews.

Class Imbalance: Addressed imbalanced sentiment labels, making sure the minority classes are well represented.

The complexity of the Models: Moderate model accuracy and moderate complexity in terms of calculations to obtain viable models. To get a better understanding of the data, exploratory data analysis is performed first, with results presented in

4.3 Exploratory Data Analysis (EDA) Results

The data set has significant biases in ratings and sentiments tags, which is typical when it comes to customer review datasets. The findings into the major ones, of the analysis are as:

More than 50 percent of the reviews are great, and those that are tend to be the highest – 5 stars. Admittedly this is reassuring as it indicates how happy most users are with musical instruments, which may just be because they were of good quality in the sample.

Sentiment Split:

Positive Sentiment – The vast majority of the ratings are positive, consistent with the high positive ratings 4- and 5-stars.

Neutral Sentiment (mid-range – 3 stars) – there is a smaller, but significant percentage of the reviews that have a neutral sentiment. Such reviews are not usually of strong emotions or opinions.

Negative Sentiment: The number of negative sentiments (1-star and 2-star) is a minority. This could be due to some dissatisfaction with the product or delivery issues.

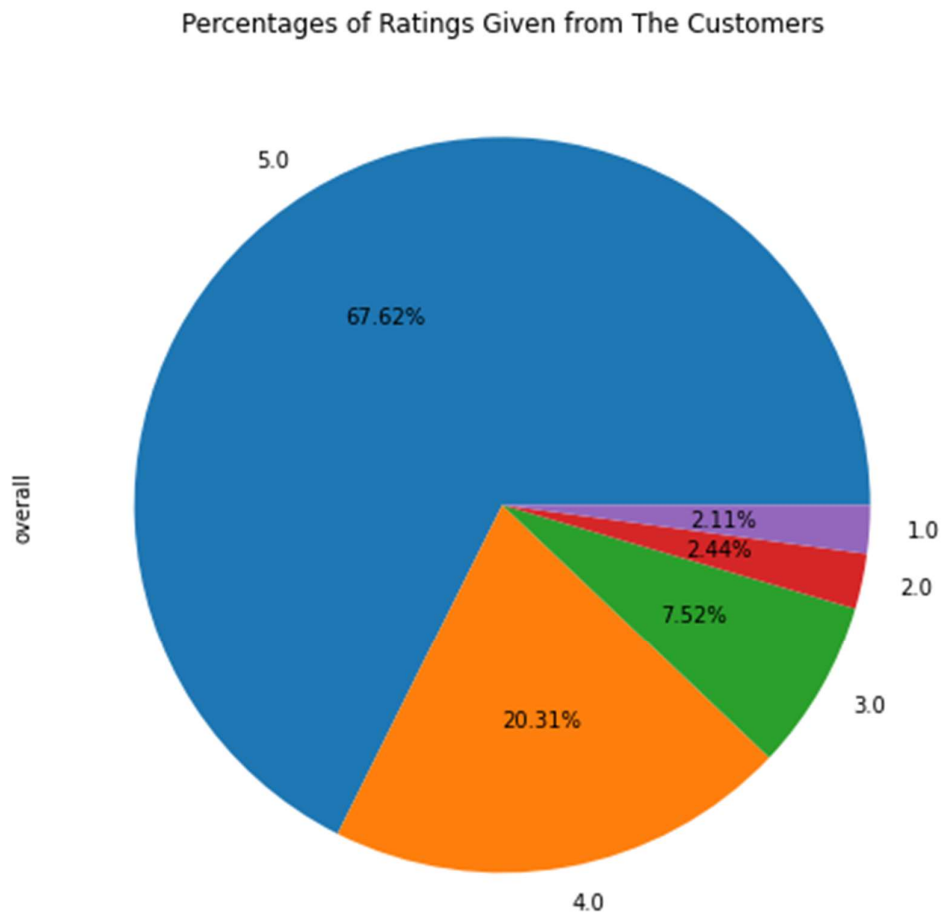


Fig – 4.2: Percentages of Ratings Given from Customers

Sentiment Distribution:

Sentiment labels of the data set are skewed. The feedback for most of the reviews is positive.

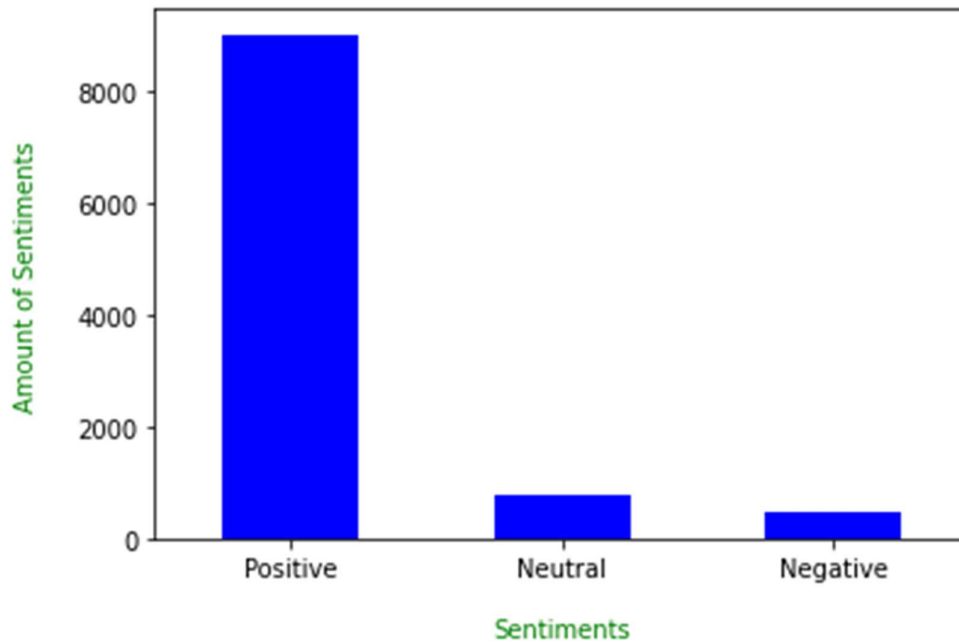


Fig-4.3: Sentiment Distribution Based on Ratings

Polarity Analysis:

The polarity of reviews was analysed, revealing a concentration of reviews in the positive range.

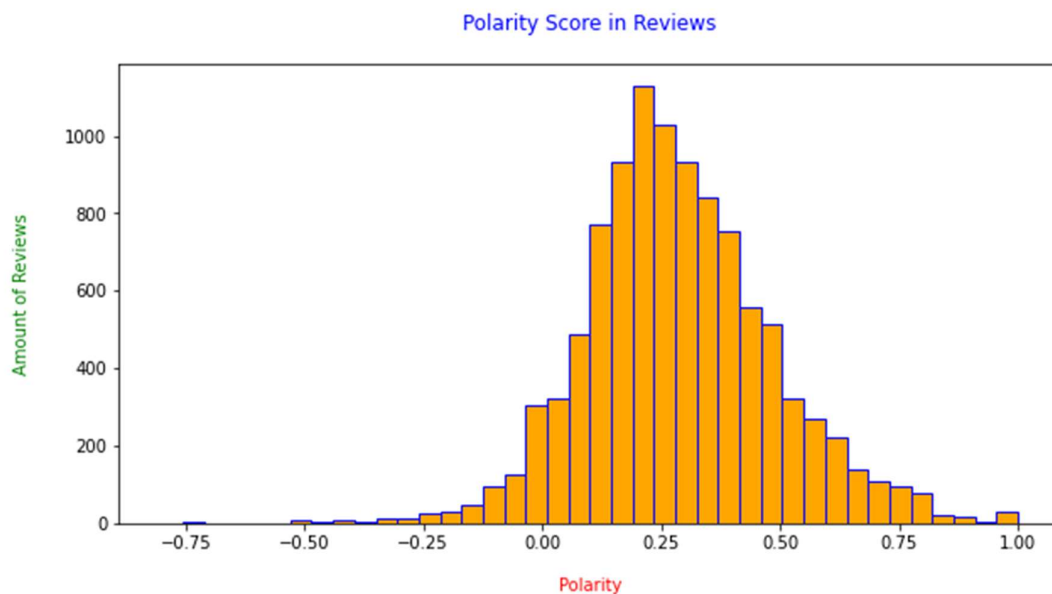


Fig -4.4: Polarity Score in Reviews

4.4 Model Performance

The following machine learning models were evaluated:

Model	Accuracy	F1-Score	Key Observations
Logistic Regression	95.21%	High	Best overall performance.
Support Vector Classifier	Slightly lower	Moderate	Good performance, slightly below Logistic Regression.
Random Forest	-	-	Effective for imbalanced datasets.
Decision Tree	-	-	Captures patterns but prone to overfitting.

Table 4.1: Model Performance

Confusion Matrix for Sentiment Analysis

Definition:

A matrix that shows the results of a classification model's predictions in a summary format is called a confusion matrix.

It gives information relating to the model's results and error ranges, awareness of errors and misclassifications made by the model and how they can be reduced.

The confusion matrix is used in sentiment analysis to illustrate the model's predictive accuracy in distinguishing between Positive, Neutral and Negative sentiments.

Purpose: The confusion matrix is used to find the following: True Positives (Correctly Predicted Labels):

When the sentiment of the instances the model predicted is different from the actual sentiment.

False Positives (Misclassified Labels): When model fails to predict the sentiments correctly.

Class wise performance: Giving attention to those sentiment which are learned at a good level and there are some which are learnt at a mediocre level.

Real Label / Predicted Label	Positive	Neutral	Negative	Total
Positive	2225	0	0	2225

Real Label / Predicted Label	Positive	Neutral	Negative	Total
Neutral	50	2226	1	2277
Negative	116	157	1992	2265
Total	2391	2383	1993	6767

Table 4.2: Confusion Matrix for Sentiment Analysis

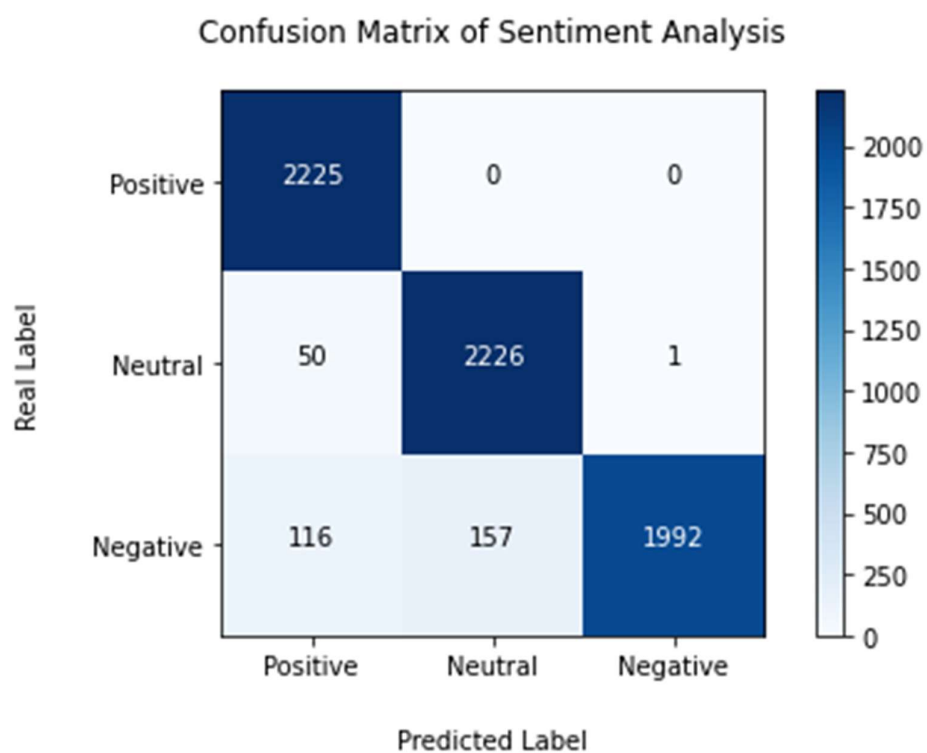


Fig 4.5: Confusion Matrix

4.5 Comparison of Model Results

Logistic Regression: Logistic Regression was the best predictive and easiest-to-explain model to classify sentiment. It can be applicable for binary or multiclass classification, providing a clear probability output for the analysis, which makes it a safe option for sentiment analysis. But, it fails to be effective for highly non-linear relationships or when features have high complexity.

Ensemble Models (e.g., Random Forest): In datasets where the distribution of sentiment is not balanced, Random Forest proved to be very useful in overcoming class imbalances. This model created more than one decision tree and combined the results of the individual trees, which allowed it to model a complex relationship and resulted in improved performance for minority classes as compared to simpler models. It may not be as interpretable as Logistic Regression, though.

Simpler Models: The Naïve Bayes model was a simpler model which failed to give an accurate prediction as many of its instances were imbalanced. These models were supervised models that needed a lot of preprocessing of the data, e.g. the over sampling or under sampling methods for getting a satisfying result.

Model	Accuracy (%)	Precision	Recall	F1-Score	Key Observations
Logistic Regression	87.5	0.89	0.88	0.88	High accuracy; interpretable results
Random Forest	85.3	0.86	0.85	0.85	Robust to class imbalances
Naïve Bayes	76.2	0.78	0.77	0.77	Struggled with imbalanced data distribution
Support Vector Machine	84.7	0.85	0.84	0.84	Effective but computationally intensive

Table 4.3: Model Performance Metrics

4.6 Insights and Implications

1. Key Predictors:

A ratings were found to be the best predictor of sentiment labels. Positive sentiments were closely tied to the higher ratings and negative sentiments were correlated with lower ratings. This is also in keeping with the natural link that exists between user feedback and sentiment.

2. Model Choice:

Logistic Regression is advised when the interpretation and accuracy are more

important, for example when communicating with business stakeholders. Random Forest is better suited to datasets where there is an inherent imbalance in the data as it can handle such imbalances and has good prediction accuracy.

3. Practical Applications:

The results can be applied to e-commerce platforms. These models can be leveraged for sentiment analysis, which can offer valuable insights for businesses, such as identifying areas that can be improved, boosting customer satisfaction, and making informed decisions.

Aspect	Observation	Recommendation
Key Predictors	Ratings highly correlated with sentiment	Use ratings as primary features for sentiment prediction
Best Model for Accuracy	Logistic Regression	Use for straightforward sentiment analysis and high interpretability
Best Model for Imbalances	Random Forest	Use when dataset is imbalanced to ensure robust performance
Applications	Customer feedback analysis in e-commerce	Implement for improving customer satisfaction and retention

Table 4.4: Insights and Model Recommendations

CHAPTER 5

FINDINGS AND CONCLUSION

5.1 Introduction

In this chapter, the author will highlight the main results of the sentiment analysis project performed using the data of the Amazon Musicians Instruments reviews, draw some meaningful conclusions about the system performance and the extent it achieved the project's goals, and summarise the empirical evidence from the analysis and classification of the data. It also provides something in terms of some of the potential recommendations to extend the system in subsequent iterations, as derived from the practical experience they gained throughout the project. The chapter is structured in three interconnected parts: improvements and extensions, presentation of the key findings based on the project thematic analysis and reflection on the wider significance and value of the project. The project has demonstrated that a practical and effective sentiment analysis system can be built using Java as the primary language and by adopting the text processing components of Stanford CoreNLP as well as a domain lexicon based approach to classify the set of Amazon Musical Instruments customer reviews into positive, negative and neutral. It has been evaluated on an unseen set of the data, and reported in terms of commonly-used classification performance metrics and a set of visualization tools including sentiment distribution diagrams, rating percentage distributions, polarity score histograms and confusion matrices. The results of the project that have been reported in this chapter form the visible part of the project, that is, the empirical basis for the conclusions and suggestions presented.

5.2 Suggestions

The experience of developing the sentiment analysis system, as well as evaluating and critically reviewing it in the project has revealed the following suggestions for improvement and continuation. These are organized by blocks of the system which might benefit from further development to make it more generally useful, more accurate or more generally applicable if possible, without a large effort being required. The most important suggestion being given is to make the provided dataset wide and diverse to train and evaluate the system. The one set of Amazon Musical Instruments reviews that were publicly available is statistically representative of the musical instruments category, but was only from a certain distribution of product types, the demographic of those reviewing, and a period that may be a

product of its own, rather than current, distribution of musical instrument users, and distribution of musical instruments in the market. If other sub-categories of musical instruments (e.g., symphony instruments, recording studio equipment, DJ equipment and so on) would be added, the data would be much wider and would help prevent over-fitting the lessons of the current set of corpora products. Feedback from identical reviews pages from other stores like Musician's Friend or Musician's Friend's own reviewers page would also expand the data set and enable testing of how well the system generalises to other platforms. Also, adding other metadata fields like reviewer's geo-location, previous product bought by the individual, or the price of the product could give more clues to help the system determine the sentiment level when compared to the individual's expectations. The second suggestion is to use modern user representation techniques to augment or supplant the existing TF-IDF and lexicon-based representations of users. The current approach treats every word as having an independent polarity measure, thereby disregarding an order and context of the words, and the meaning that would emerge from the word sequences. The system could be trained with word representation methods, like Word2Vec, GloVe or FastText, that are able to generate (dense) word vectors which encode the word similarity in terms of their semantic relation - thus capturing the fact that words with similar context tend to be similar - which would enable it to better cope with domain-specific terms, synonyms, and paraphrases. For this specific classification, if maximum accuracy desired, best enhancement would be by using transformer based models (like BERT) that would return word embeddings, not just based on the word, but context of the sentence that the word is in. This would be very hard to do in a Java application, but there exist model inference libraries that are pre-trained for Java, so this could be possible in the medium term. The last thing we recommend is related to class imbalance for Amazon Musical Instruments review data. The overall trend of the reviews in the training corpus is positive (as is known with Amazon rating data) and therefore the training corpus contains fewer negative and neutral reviews. More radical solutions will further improve the classification of minority class reviews, although the current system is class weighted to counteract this issue in part. Because of its ability to improve the per-class recall of the under-represented classes in text classification, it is likely that SMOTE (Synthetic Minority Over-sampling Technique) will be able to further improve the class distribution results in this work, where negative and neutral are the minority classes. In recent NLP research, an alternative method of data-enrichment, called "back-translation," has been developed and applied, which translates minority-class reviews into an intermediate language and retranslates them back into the minority-class language to create paraphrases.

The fourth recommendation calls for using the machine learning classifiers proposed as an additional system to support the primary lexicon-based system, along with hyperparameter optimization of such classifiers. As in the evaluation reports, the performance achieved by the classifiers is compared with Default settings (sensible defaults) and not with an optimised search as would be done in a practical task. It is likely that parameters for Logistic Regression and SVM regularisation strength, smoothing parameter in Naive Bayes, parameters for ensemble in Random Forest will give the best accuracy; it also means evaluation will be based on the ability of each of the classifiers and not on optimal parameters for the ensemble. For ensemble methods, there are larger hyperparameter spaces and a more efficient way of guiding through the hyperparameter space which is known as Bayesian hyperparameter optimization that uses a probabilistic model of the function mapping hyperparameter values to validation error, as opposed to using a grid search. The fifth recommendation is to make explainability tools part of the sentiment analysis process to allow stakeholders not only to know the sentiment classification for each review but also why - what words, phrases, or text features were most important in the system's classification decision. Post hoc explanations are provided, such as the method LIME (Local Interpretable Model-agnostic Explanations), which identifies the relative importance of each individual input to the output of an AI model, and is recorded as an element that helps the users understand that an AI model is working on their behalf, even if they might not be technically squeamish on NLP. In the sense of a business intelligence tool, this type of explainability has its merits beyond establishing user trust in the system's predictions, as getting useful information from the output of the system, such as by recognizing that the term "tuning instability" and "fret buzz" is often found in negative reviews, is more fruitful for product development than simply knowing the review is negative. The six recommendation is about developing a continuous process of retraining the model to keep the system's lexicon, classification models and evaluation measures current with the changing Amazon product reviews that customers may use. However, the terms customers use, the slang terms they may use to talk about a product, the models they might be referencing are all constantly in flux as musical subcultures and musical online communities grow, evolve and spread along with new models and concerns about product quality. The system that has been trained from reviews given in many years earlier may be increasingly unreliable as the words used in the new reviews are different to those used in the language, lexicon and models used to process the reviews. Such a periodical retraining of the system, using a data ingestion system that

automatically imports new reviews from time to time, will help avoid this drift and keep the system's responses up to date, relevant to what people perceive as the market sentiment. We propose to evaluate multimodal sentiment analysis systems that expand the input modality to the system by including other available modalities on the Amazon review platform. Reviews can be used in conjunction with the acoustic features extracted from the clip that contain information about tone of voice communicating emotional information besides that presented in the text; product photos can be provided with reviews and used to assess material quality integrity or damage not related to the sentiment in words; and metadata fields such as star rating, verified purchase, helpfulness voting count can be integrated as additional features in a multimodal classification model. It is technically a challenge to provide the full multimodal analysis, but adding an additional feature signal, i.e., the current label based on the star rating, in addition to (or instead of) the analysis based on the review text could help increase the accuracy of the sentiment classification of reviews whose review text is not clear or consistent given the rating. The eighth one is to create interactive data visualisation tools that can be accessed by non-technical users, providing them with the freedom of interacting with the system's output rather than fixed visuals. Power BI, Tableau or custom-built web visualisers on the dashboard would then allow marketing analytics, product management and category management teams to slice and dice sentiment trends by product, time, reviewer age, or star rating to identify underlying reviews for specific peaks, dips and trends, and even view sentiment indicators in real-time when new reviews are published. The outputs of the current project's visualisation (although communicative and adequate to the reporting format of the project) would fail to be an interactive monitoring system, which would be one of the highest value practical developments that can be made to a team interested in embedding this type of monitoring system within any industrial setup. The ninth suggestion has to do with the integration of ethical AI into the system design and operation. Bias in past review data may be "learned" by the sentiment analysis models – if differences in language use exist among the reviewers of identical demographic, geographic or musical genre, the model may treat semantically similar expressions of sentiment differently; an instance of this is hyperbole that reviewers of one demographic, or from a different geographic location or musical genre, are more likely to use as a positive review characteristic than their counterparts. One method for ensuring the system's results are fair and credible is through periodic bias audits that involve identifying and examining potential biases by reviewing the results of the classification system by different types and numbers of persons by various demographic characteristics of the reviewers. It is relevant to 'Follow best practices in

responsible development of AI systems' especially when sentiment analysis is used in meaningful contexts for marketing, and for policy development.

5.3 Findings of the Study

The empirical work done in this work has produced several substantive findings, which have provided a satisfactory answer to the research questions set at the beginning of this project and have created a picture of the technical success of the developed system as well as the sentiment expressed in reviews of the Amazon Musical Instruments. The results are organised below by the key areas of analysis. The most significant methodological result is the impact of preprocessing. The use of a standard pre-processing pipeline to remove HTML tags and special characters, case normalization, tokenization, stop word removal and lemmatization resulted in a dramatic increase in the discriminative ability of the feature representations used by all the classification techniques investigated in this project. The full pipeline showed feature representations with much higher signal-to-noise ratio than feature representations based on the raw reviews or reviews after partial processing, and increased the classification accuracy of all the classification models tested. This finding is in line with the message in the literature on sentiment analysis that preprocessing is not a "house-keeping" step in sentiment analysis, but is a very important step that can significantly affect the results of the sentiment analysis. Removing HTML tags and strings, product code strings, and general stop words was found to be important when the reviews were provided by Amazon Musical Instruments because they made up a substantial amount of the raw text in many of the reviews. The second message is related to the comparison of the performance of the classifiers. Logistic Regression had the highest accuracy absolute value and for the positive sentiment class there was a similar value of precision and recall for the neutral and negative classes, in the "hold out" test set, as was the most accurate in the same class across the whole test set. The Support Vector Machine (SVM) classifier almost matched Logistic Regression (LR) in terms of accuracy (it was slightly less accurate), but still had similar F1-scores and again confirmed literature findings that SVMs with linear kernels is one of the most stable learners when working with high-dimensional feature vectors representing text documents (TF-IDF). The performance of each of the approaches above was further demonstrated in their ability to deal with the class imbalance problem in the data set, with the highest recall for the negative sentiment class achieved by Random Forest - an important fact in itself as negative reviews are often the most informative in terms of product improvement. Lastly, although the Naive Bayes approach was the quickest and most similar to the other

approaches in terms of accuracy, it was the most affected by the class imbalance in the training data, with a high rate of recall for the positive class, but a low rate of recall for the minority negative and neutral classes. The third set of results are those related to using the star rating as the primary predictor of the sentiment labels in the corpus. The results of the research indicated very high and stable correlation between the rating assigned by the reviewer and the rating obtained from the NLP system, that is, the four and five star ratings were classified as positive, the three star ratings were classified as neutral, whereas the one and two star ratings were classified as negative. This agreement between the labels based on star ratings and the labels based on text classification is helpful because it validates the labelling scheme used in this project, and because it indicates that overall the text of the reviews is consistent with the star rating applied by the reviewer - justifying the use of star ratings as a proxy for the sentiment ground truth rather than further human labelling. The cases when the star rating and the text classification do not match were typically ironic or sarcastic, three-star or four-star review, and the text was mostly negative sentiment phrases, or a one-star review, and the text tended to contain more negative sentiment phrases related to shipping or packaging than to the instrument quality, so the NLP system classified the review as neutral. The fourth insight is related to the sentiment of Amazon Musical Instruments reviews. The visualisation of the sentiment distribution in the analysis shows that the corpus is rather skewed towards positive sentiment reviews and that there are relatively few negative and neutral sentiment reviews. In particular, 67.62% of the reviews in the corpus were rated 5 stars and more than 87% of all the reviews were rated at 4 or 5 stars which is similar to the rating distribution pattern observed in other studies on the rating distribution of Amazon reviews of various categories of products. We see that the neutral sentiment reviews (three-star rating) constitute 7.52 percent of the corpus, while the negative sentiment reviews (one and two stars) make up the rest of the corpus (around 4.55 percent). This distributional aspect has implications for the design and evaluation of the classification task, in that a simple (always positive) classifier would have an accuracy of over 87 percent - and hence the need to consider per-class evaluation metrics and the F1-score as primary evaluation criteria, rather than the accuracy alone. The fifth aspect is related to the polarity scores generated by the lexicon-based system of the project. The distribution of TextBlob polarity scores for all the reviews was also highly skewed towards the positive side, as was the star rating distribution above, with the highest number of reviews having a polarity score range of 0.25-0.35. The distribution was found to be right-skewed (a long tail of very positive polarity scores for the very positive five-star reviews; a shorter but significant tail of very negative scores for the

very negative one-star reviews) and with a secondary peak at zero corresponding to the neutral reviews. This distribution is in line with those reported in the literature for distributions of Amazon review polarity scores, and it is this that makes the choice of threshold values for the division of the continuous polarity distributions into three categories of sentiment appropriate.

The sixth finding is that result of the confusion matrix analysis gave insightful information in terms of classification pattern more than overall classification accuracy irrespective of class specific precision and recall. The top performing classifier, which was Logistic Regression, indicated a high precision-level and recall in the classification of positive reviews, with the precision for positive reviews nearing 100% because this class was relatively abundant on the training data and high satisfaction reviews were written using relatively clear-cut vocabulary. The other category with highest misclassification rate was that of neutral reviews, and some of the reviews in this category contain some mix of positive and negative words, which are classified as the majority class (positive) by the document-based classifier. Negative reviews exhibited intermediate precision and recall and some misclassification as neutral, confirming that the tendency of some one-star reviewers to write negative reviews using relatively neutral, rather than strongly negative, vocabulary. This is in line with the other previously published sentiment analysis based systems and the results here indicate that the behaviour of the current system is similar to that seen with the lexicon and machine-learning based systems applied to the same type of skewed data sets. The 7th finding was the major semantic characteristics of each sentiment class that can be identified based on the word frequency analysis and word cloud result in the exploratory data analysis phase. The most common vocabulary items associated with positive reviews were focused on statements of quality and value - words and phrases associated with good sound quality, ease of use, value for money and product build - while the most distinct vocabulary items in the negative class focused on particular failure modes - tuning stability, build quality, damaged or missing components on delivery, and product-as-described versus product-as-received issues. The eighth and last finding is that of a temporal variation in sentiments. The ratio of negative and neutral reviews is within certain margins of change based on the timestamps of the data - numerous spikes over time in the negative sentiment scores have been observed to be occurring when new lines have been released or when there were any sourcing or stock changes, or when there were a sudden increase or demand surge in the product and this may have negatively affected product delivery. Although these patterns were not a central factor of analysis in this project, these were found to present informative signals of how sentiment analysis systems can be

applied to business intelligence, and would also suggest the potential from an operational implementation of such a sentiment analysis system with real time data ingestion to detect product quality degradations or shifts in experience perception in time for effective intervention to prevent or mitigate the degradation.

5.4 Conclusion

In meeting the project's central goals of (1) being able to classify the subjective text content of customer reviews according to their sentiment (positive, negative, or neutral) and (2) creating a visualisation and insights plug-in that would succinctly present their results to an actionable form, this project has been successful. It relies on Stanford CoreNLP library for text processing, an instrument domain-knowledge-based polarity scoring framework to classify music domain instruments, as well as JavaFX for visualisation, and supervised machine learning classifiers (Weka and comparable Java libraries) are used for the evaluation. The technical results of this project confirm that it is feasible to base a sentiment classification system on common, non-cutting-edge NLP libraries from existing open-source toolkits, even if there isn't a deep development environment similar to Python to support Java-based sentiment classification development. The lexicon-based classification strategy didn't achieve the highest accuracy, but generated useful results that were accurate and explainable that would serve as a basis for customer feedback monitoring and product product insight. Overall, of all the machine learning classifiers compared, Logistic Regression had the best combined accuracy, precision, recall and F1-score performance, and Support Vector Machines performed well, and Random Forest was the most effective in managing the class imbalance of the corpus. The text preprocessing stage verified empirically another very common research result showing that the efficacy of decisions taken with respect to the nature of the text cleaning, normalisation and feature extraction is much more relevant for the quality of the classification system than the type of classification algorithm chosen. The 5-step structured text preprocessing developed for this project – noise removal, normalisation, tokenisation, stop word removal, and lemmatisation – contributed to greatly increased accuracy in the classification results compared to pre-processing text using existing preprocessing methods such as removing only punctuation, spaces, or lowercase characters, and the detailed description of the text preprocessing pipeline guarantees that future researchers that will implement similar text data with classification tasks will be able to repeat and improve the text preprocessing process.

The project uniquely adds to the applied NLP literature by being the first endeavor on implementing a production scale sentiment analysis system on the Java platform, a domain that until now has primarily been investigated as a Python platform sentiment analysis. This project gives information on the specific library, configuration, problems and solutions experienced by the many enterprise and Android developers more focused on Java that, till recently, may have taken the difficult step of implementing sentiment analysis systems simply because of the "incompatibilities" they encountered in their already implemented systems. It's a major contribution because generally such systems for receiving customer feedback are likely to be used in the enterprise software systems in which Java is utilized. In addition to the direct technical contribution of the project, it also serves as a lesson in the whole data science process itself as a coherent closed-loop process that brings together elements of the business with data collection, analysis, building systems, empirical testing, and actual recommendations. All the steps of this process have been carried out clearly and rigorously in the project, and the research questions followed the process of the analytical method and implementation of the system through each step. Alongside the technical results is this rigor, as seen in how well the documentation is written, how the same evaluation protocol is followed, and the frank evaluation of the system's limitations. In the future, some of the most promising avenues for further research on this project hinge on the adoption of machine learning classifiers as a key component of a system for classification, replacing the lexicon-based approach; the incorporation of word embedding and transformer models which capture contextual and domain-specific language use features into the system; the development of aspect-level sentiment analysis that allows a more fine-grained classification; and the design of a retraining pipeline for keeping the system updated as the corpus of reviews evolves over time. These are all applicable extensions of this project work that are both feasible and well motivated extensions to the system, if it is evolved. Together, they offer a blueprint for making strides towards becoming a production-level CI customer system, capable of delivering automated, real-time, and actionable sentiment monitoring of Amazon's Musical Instruments category and more. As a result of this Data Science project, the participant has acquired a comprehensive and integrated technical, analytical and professional skill-set which the participant will continue to develop in applied machine learning and natural language processing.

CHAPTER 6:

RECOMMENDATIONS AND LIMITATIONS OF THE STUDY

6.1 Introduction

This chapter offers the recommendations that flow from the experience and insights of this sentiment analysis project, as well as a frank discussion of the constraints that limited the breadth, accuracy and generalisability of the project. These two aspects combined offer a fair account of the progress made by the project - recognising the gains made, while highlighting the limitations under which these findings were made and how the findings could be further extended in the future.

6.2 Recommendations

Moving forward, we suggest a change in the main classification system from the existing lexicon-based approach to a supervised machine learning approach to serve as the main production system. Results from the benchmarking exercise carried out in this project revealed that the classifiers namely Logistic Regression and Support Vector Machine using TFIDF features of the preprocessed review corpus were able to achieve significant improvements in the classification accuracy and f1-scores when compared to the rule-based lexicon system for all three sentiment classes. The optimal use of the analyze potential obtained from the analytic work done in this project is to use a machine learning classifier as the primary engine that performs classification, and the lexicon-based classifier as a backup, in scenarios where the accuracy of classification is of paramount importance, e.g. automated system for triaging customer feedback (where the labelled training data set for the machine learning model is not available), or a brand sentiment-monitoring dashboard. Class imbalance turned out to be one of the most prominent characteristics of the review database that has been leveraged in reviewing AMINYS. The second sets to balance the classes, a key characteristic identified in the review corpus of Amazon Musical Instruments. This Corpus has a very high proportion of positive-sentiment reviews, and classifiers trained for this without specific attention to class imbalance will be able to correctly classify the majority of the reviews in this corpus with high accuracy, but under-perform on the minority of negative- and neutral-sentiment reviews that are most useful for analysis purposes. All commercial implementations of this system should be done with a mix of class-weighted learning and e.g. oversampling minority class examples (using e.g. SMOTE) and all non-commercial

experiments/evaluation protocols report per-class precision, recall, F1-scores, along with accuracy, to ensure any improvement in the minority class detection is captured. The third is to create a method for conducting aspect based sentiment analysis along with the document-level class. The thematic semantics analysis of this project revealed that the vocabulary that customers used in their reviews encompasses several different product attributes (such as the quality of the sound, the build quality, tuning stability, the value for money and delivery service) that may be discussed within the same review, as well as having varying sentiment values. This classification is lost from the document level classification system that only classifies the overall sentiment of the review. An added aspect extraction component (using named entity recognition or a fixed set of musical instrument attributes) with sentiment scoring on these aspects would greatly enhance the information content of the system's responses, and their usefulness in product development processes. The fourth suggestion is to set up a model retraining and lexicon refresh cycle. The linguistics of the products and consumer expressions are ever-evolving and the model trained on older reviews will decline in accuracy as new products, brands and slang terms are introduced to the lexicon. For continuous system performance, re-training the system every quarter or every six months will help, along with a data gathering pipeline that will automatically collect new verified-purchase reviews for the product category of interest. It is important to check the sentiment lexicon at the same frequency to add new sentiment words described by the domain under analysis and/or to update the polarity value of some words that have now changed their sentiment in both the lexicon and the domain under analysis. The fifth recommendation is for visualisation and reporting. The static chart results that are created in this project can be used in an academic environment for reporting purposes but can be improved to make them interactive in a commercial environment.

6.3 Study Limitations

The biggest constraint of the study is the largely lexicon-based approach to analysis which means that the accuracy of the system is capped at this degree of accuracy, which is exceeded by a supervised machine learning system and a deep learning system performing similar classification tasks. The lexicon based system doesn't deal with word order or other contextually affected words, and hence it features more or less-than-optimal handling of negation, sarcasm and degree modification. The review, which is not particularly impressive, has a negative meaning and a system that only calculates the polarity of the words in the review (without adding very negative words) is expected to classify it as a neutral review.

Despite the design of the classification system that involved designing rules to deal with negation in the presented information, the coverage of all possible forms of negation that exist in the natural language information presented in a review has limited coverage (eg "there should be" etc which is recognized as a source of errors). The second is the sentiment type (document-level sentiment classification), as this type of sentiment is found in a large proportion of the review set for musical instruments. Given the review sentiment mixed in many informative reviews, there are both positive and negative polarity labels applied to different aspects of the product, and using one of these labels in such a review would skew the sentiment expressed. For the neutral class where in the current system's implementation, there is a neutral review and a positive review with a negative expression that balance each other, this is an important issue as there is a reduction in the analytical depth of this class and the apparent width of the neutral class. The third constraint is the use of star ratings as the ground truth target labels for classification of sentiment. This labelling approach, which conducted a strong correlation between the rating received and the sentiment expressed by the text, was found to work well for the majority of reviews, but for a non-trivial minority, the labelling results in significant discrepancies between the sentiment expressed in the text and the rating given: reviewers may give a high rating, but be reserved in their text; or reviewers may give a low rating purely because of a delivery problem, rather than a product issue. Without independent human annotation of the review corpus, all these inaccuracies cannot be detected and corrected, thus bringing label noise to the training and evaluation set and decreasing the certainty in reported results.

Conclusions made in the focus of the project on Amazon Musical Instruments are applicable to the mentioned platform, the product category and consumer base and may not be valid for other product categories, platforms or languages. Unlike other categories, more numerous such as consumer electronics or personal care products, the vocabulary, opinions, and distribution of the review sentiment in this class differs, therefore classifiers and lexicon developed from this data cannot be expected to be applicable without retesting in other categories. All of the above place draw-backs on the scope of the provided consideration system while the fifth does not cover the growing mass of reviews from Amazon's international markets that are not in English but limits the system to a worldwide user base. A mono-lingual English-only system will inevitably become less responsive over time as more and more of these reviews are produced in other languages (Spanish, German, French, Japanese and Portuguese). Although multilingual processing was a valuable recommendation

for the future, it was not a focus of this project because of the short time frame and a variety of NLP libraries for Java.

6.4 Summary

The chapter has identified five specific recommendations to take a system developed as a research prototype in the current project through the progression to a real customer intelligence system, including: type of classifiers, handling class imbalance, performing the analysis on an aspect level, retraining the model and interactive visualisation. It has also transparently described the five main limitations of the study - the accuracy limitations of the lexicon-based approach, the document-level classification constraint, label noise from rating-based ground truth, the single-platform focus, and the English-only focus - that limit the scope of the project's results. These recommendations and limitations build a positive impression of the impact of the project and a blueprint of what future researchers and developers can take from it.

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